Beyond Spectral Shift Mitigation: Knowledge Swap Net for Cross-Domain Few-Shot Hyperspectral Image Classification

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***Abstract*— Spectral shifts between source and target domains (TDs) pose significant challenges in cross-domain hyperspectral image classification (HSIC). Current methods often struggle to balance mitigating these shifts while preserving crucial TD information, which limits their ability to leverage spectral priors and domain-specific characteristics for accurate classification. Our work proposes a novel knowledge swap net (KSN) for few-shot cross-domain HSIC. KSN tackles the challenge by enabling effective knowledge transfer between homogeneous (spectral) and heterogeneous (domain-specific) feature spaces through a two-step knowledge swap strategy: leveraging homogeneous knowledge distillation (Homo-KD) for transferring spectral knowledge and heterogeneous meta-learning (Hetero-ML) for model refinement with TD feedback. In addition, we develop a relative distance difference (RDD) loss function to improve feature discriminability under few-shot conditions. Experiments conducted on four target datasets demonstrate the superiority of KSN. Notably, KSN achieves a remarkable overall accuracy (OA) of 82.56% on the Houston University (HU) 2013 dataset, surpassing other leading methods by 3.83%–8.98%. The source code will be available online: https://github.com/ZhaohuiXue/KSN.**

***Index Terms*— Cross-domain, few-shot learning (FSL), hyperspectral image (HSI), knowledge distillation (KD), meta-learning.**

I. Introduction

HYPERSPECTRAL imaging has significantly advanced remote sensing technology by capturing intricate details across a broad spectral range [1], [2], [3]. Noted for their precision in distinguishing similar materials, hyperspectral images (HSIs) are utilized in various fields, including precision agriculture [4], mineral exploitation [5], and environmental assessment [6], [7]. These diverse applications underscore the importance of HSI classification (HSIC)—the process of assigning specific labels to each pixel vector [8].

The advent of deep learning (DL) has expanded HSIC methodologies, with techniques based on various neural network architectures, including recurrent neural networks, convolutional neural networks, graph convolutional networks, and transformers, being extensively proposed [9], [10], [11], [12]. However, high-performance HSIC remains challenging due to the heavy reliance on labeled samples, a dependency that creates efficiency and practical applicability bottlenecks [13]. The reliance also limits the adaptability of HSIC in dynamic environments where obtaining extensive labeled data is impractical. Consequently, there is a growing interest in methodologies that reduce reliance on labeled samples, forming a promising trend in the field.

The pursuit of high-precision HSIC with limited samples, known as few-shot learning (FSL), has led to the development of metric learning, active learning, semi-supervised learning, and transfer learning (TL). Metric learning approaches like episodic training [14], [15] and contrastive learning [16], [17], [18] have proven effective in handling few-shot scenarios. Recent work also incorporates multiview learning [19] and generative adversarial learning [20], fostering more varied training data to mitigate overfitting. Active learning offers a straightforward approach by allowing manual selection of uncertain samples [21] but suffers from being time-consuming and expensive. Semi-supervised learning [22], [23], [24], which optimizes models using numerous unlabeled samples, still faces challenges with sample selection bias [25]. To recap, methods confined to single-domain samples face considerable challenges in transcending these limitations. Conversely, leveraging cross-domain samples offers potential for further improvements, thereby drawing significant attention to TL.

TL offers an alternative by leveraging knowledge from a well-labeled, data-rich source domain (SD) to improve classification accuracy in a data-scarce target domain (TD) [26]. According to Pan and Yang [27], we categorize TL methods in HSIC into four types as follows.

1) *Weight/Priori Transfer-Based TL:* Leverages pretrained model parameters to reduce reliance on labeled data in TD [28], [29].
2) *Sample Reweight-Based TL:* Trains the target model with reweighted SD data, prioritizing samples more relevant to the TD [30], [31].
3) *Feature Alignment-Based TL:* Aligns source and target data feature to reduce domain discrepancy, thus enhancing model applicability [32], [33].
4) *Relation Knowledge-Based TL:* Utilizes stable interdomain relational knowledge for more accurate feature mapping across domains, incorporating strategies like graph embedding [34] and cross-modal mapping [35].

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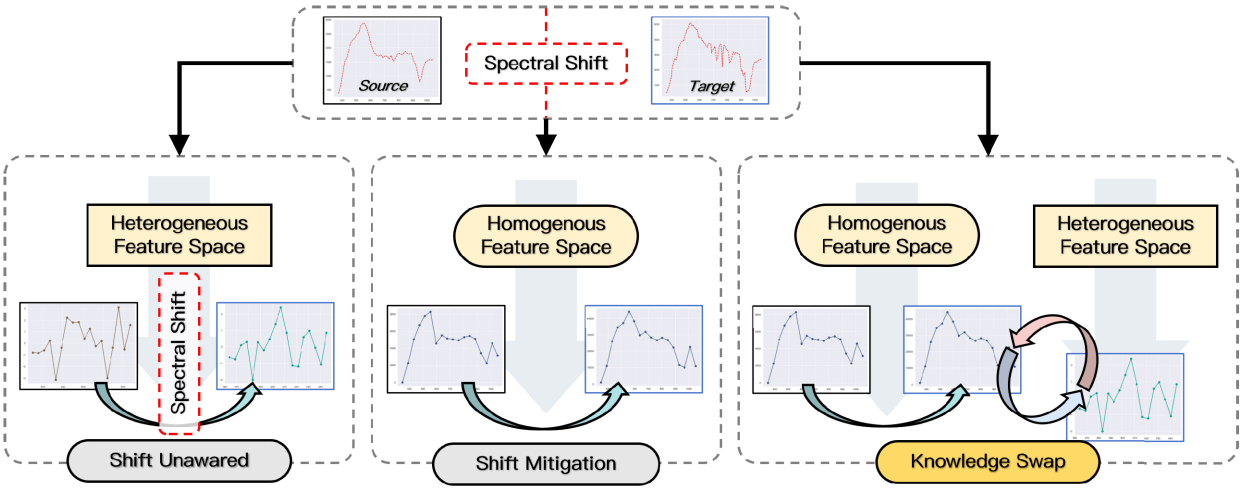


Fig. 1. Comparison of approaches to addressing the spectral shift in cross-domain few-shot HSIC. 1) Shift-unaware methods overlook the alignment of physically meaningful data inherent in spectral features. 2) Shift mitigation methods neglect the loss of TD-specific information. 3) The proposed knowledge swap method mitigates spectral shifts across domains while preserving the unique characteristics of the TD.

Although effective, traditional TL methods often struggle when adapting to target HSIC tasks that significantly deviate from the SD, especially under data scarcity. These challenges arise from TL's reliance on assumptions about data distribution, which may not hold true across domains with varying spectral characteristics [36]. For instance, spectral shifts caused by differences in sensor types, atmospheric conditions, or illumination can lead to misalignments between the source and target feature spaces, hindering the effectiveness of TL.

To address these limitations, researchers have developed cross-domain FSL (CD-FSL) methods [37], [38], aiming to reduce the requirements for data consistency across domains and the labeling efforts needed in the TD. Concretely, CD-FSL enhances traditional TL by explicitly bridging feature space, feature distribution, and task-related disparities between domains.

1) To resolve feature space discrepancies, often stemming from varied spectral ranges or resolutions, techniques such as domain generalization [39], [40] and subspace alignment [41], [42] are employed to make the feature representation more comparable.
2) To address feature distribution variations, primarily due to different illumination conditions or acquisition locations, strategies like distribution alignment [31], [43] and adversarial training [36], [38], [44] are commonly utilized.
3) When dealing with distinct classification tasks across domains, adapting model parameters proves advantageous. Effective methods include: model-agnostic meta-learning (MAML) [45] trains models to find optimal parameter initializations, which accelerates adaptation to new tasks. Knowledge distillation (KD) [46] involves transferring insights from a complex model to a simpler one, boosting the later's capability in the TD. In addition, multitask learning [47] enables models to learn multiple related tasks simultaneously, thereby developing a broader ability to generalize across new, related domains.

Despite achieving notable classification accuracy, existing methods often overlook aligning physically meaningful data inherent in spectral features during TL. Such oversight results in inaccurate transfers or misleading information, which could hinder performance in TD. For instance, [41], [42], and [48] standardize the dimensions of HSIs through domain-independent mappings, such as band selection or feature transformation, resulting in an equal number of channels but with mismatched heterogeneous feature space. To address the problem, [49], [50], and [51] have proposed alignment-based methods to mitigate spectral shift between source and TDs. However, these methods often neglect the loss of TD-specific information during feature homogenization, compromising the discriminability of TD features (see Fig. 1).

To address these challenges, we propose the knowledge swap net (KSN), a novel model that mitigates spectral shifts across domains while preserving the unique characteristics of the TD, thus maintaining and potentially enhancing the discriminability of features. First, the spectral response convolution (SRC) aligns spectral data from different domains into a homogeneous feature space, ensuring spectral integrity. Meanwhile, principal component analysis (PCA) is used to map TD data into a heterogeneous feature space, capturing domain-specific information beyond spectral characteristics. Second, the proposed spatial–spectral residual network (S2RN) functions as the backbone. S2RN builds upon the strengths of the spatial–spectral network (SSN) [52] by incorporating an enhanced residual connection and a relative distance difference (RDD) loss to boost feature discriminability. Third, the knowledge swap strategy employs homogeneous KD (Homo-KD) and heterogeneous meta-learning (Hetero-ML) to facilitate efficient and bidirectional information exchange across feature spaces. Homo-KD transfers SD knowledge to the heterogeneous model, while Hetero-ML provides feedback from the heterogeneous model to the homogeneous model, facilitating mutual enhancement and consistent model improvement.

The main contributions of our work are summarized as follows.

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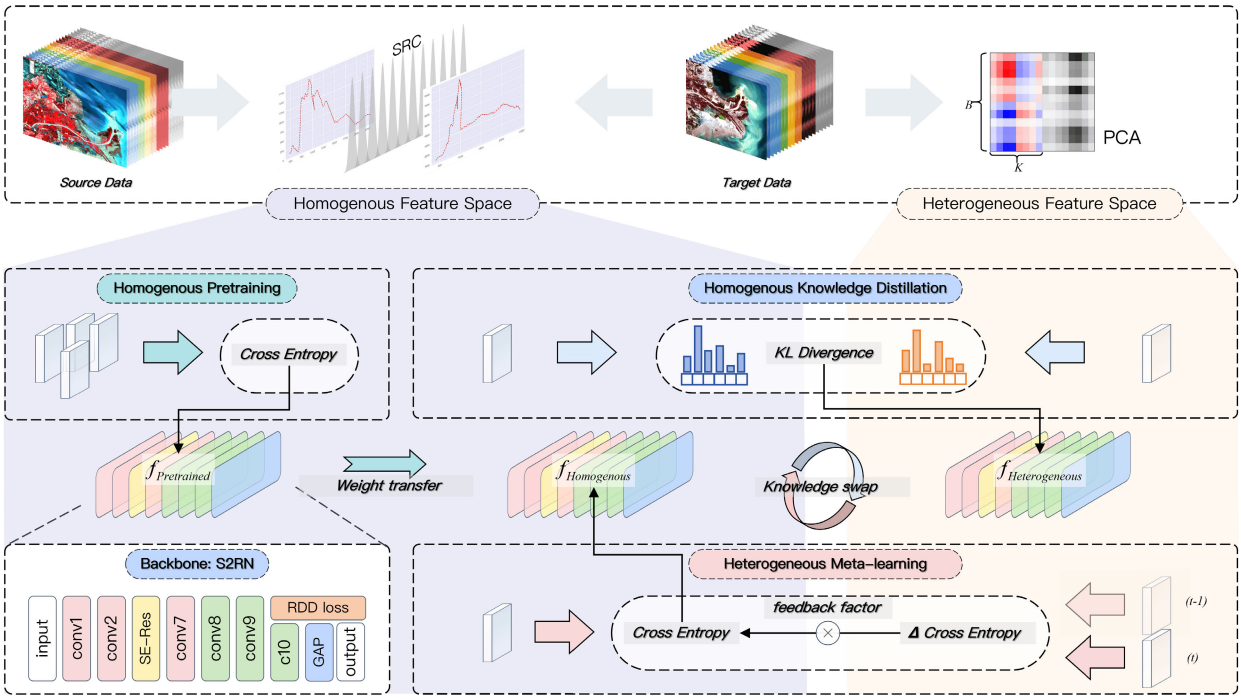


Fig. 2. Overview of the proposed KSN. First, SRC and PCA construct homogeneous (spectral) and heterogeneous (domain-specific) feature spaces. Second, the proposed S2RN, integrated with SE-Res blocks and RDD loss, functions as the backbone. Third, a pretrained homogeneous S2RN model transfers knowledge to the heterogeneous model (Homo-KD). Hetero-ML then guides the homogeneous model to improve using feedback from the heterogeneous model, enabling knowledge exchange across feature spaces.

1) A novel KSN employs a knowledge swap strategy across homogeneous and heterogeneous feature spaces has been developed. By incorporating Homo-KD and Heterogeneous ML, KSN facilitates efficient and dynamic knowledge swaps between source and TDs, significantly enhancing classification performance in HSIC CD-FSL scenarios.
2) We show that SRC effectively mitigates spectral shifts in cross-domain HSIC tasks. Experimental results within the KSN framework indicate that SRC facilities knowledge transfer from the pretrained model.
3) A robust network backbone, S2RN, has been developed and enhanced to more effectively extract informative features from HSIs. Specifically targeting FSL scenarios, we have designed an RDD loss to enhance feature discriminability.

II. Related Work

A. Knowledge Distillation

KD enables the transfer of knowledge from a complex "teacher" model to a simpler "student" model. Within the teacher–student (T-S) framework, the student model learns to replicate the output of the teacher, known as pseudo-labels, by minimizing the Kullback–Leibler (KL) divergence between their outputs

$$\ell_{\mathrm{KL}}\left(f_S, x, \hat{y}\right) = \sum_x \hat{y} \log \frac{\hat{y}}{f_S(x)}. \tag{1}$$

KD utilizes a temperature coefficient (τ) to moderate label smoothness, thereby reducing overfitting risks and enhancing interclass pattern transfer

$$\sigma_\tau(z_i) = \frac{\exp(z_i/\tau)}{\sum_j \exp\left(z_j/\tau\right)}. \tag{2}$$

In this study, we employ KD to convey model insights from a homogeneous to a heterogeneous feature space.

B. Model-Agnostic Meta-Learning

MAML optimizes models across multiple tasks using gradient descent to quickly adapt to new challenges with limited data. The process involves iterative parameter adjustments

$$\theta_i = \theta - \eta \nabla_\theta \ell_{\mathcal{T}_i}(f_\theta) \tag{3}$$

$$\theta^* = \arg\min_\theta \sum_{\mathcal{T}_i \in \mathcal{T}} \ell_{\mathcal{T}_i}\left(f_{\theta_i}\right) \tag{4}$$

where η is the learning rate, $\mathcal{T}$ represents a distribution over tasks, $\mathcal{T}_i$ is a specific task sampled from $\mathcal{T}$, $\mathcal{L}_{\mathcal{T}_i}$ is the loss function for $\mathcal{T}_i$, θ and θ_i are the model parameters preadaptation and postadaptation to $\mathcal{T}_i$, respectively.

In this work, we apply MAML to facilitate cross-feature optimization, allowing the homogeneous model to improve through feedback from the heterogeneous model.

III. Methodology

As shown in Fig. 2, the proposed approach mainly includes the following components: homogeneous and heterogeneous feature space construction, S2RN backbone, knowledge swap via Homo-KD, and Hetero-ML.

A. Spectral Response Convolution

To address the spectral shift between the SD and TD, we introduce SRC. This method constructs a homogeneous feature space, $\mathcal{X}_h$, by convolving the original data with simulated spectral response functions (SRF). Fig. 3 illustrates the SRC process.

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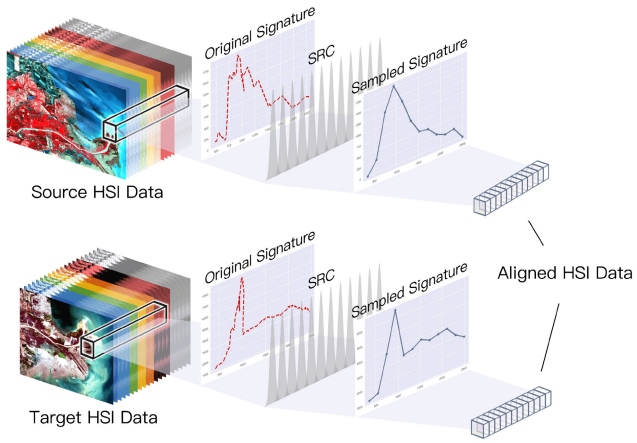


Fig. 3. Illustration of aligning HSIs with different spectral characteristics through SRC.

1) Determination of Homogeneous Feature Space: The homogeneous feature space is delineated by determining the central wavelength for each of its dimensions. To accomplish this, the shared spectral range ($\lambda_{\text{start}}, \lambda_{\text{end}}$) between SD and TD is first identified. Then, the central wavelength for each dimension is calculated based on the preserved band number B

$$\lambda_i = \lambda_{\text{start}} + (i-1) \times \frac{\lambda_{\text{end}} - \lambda_{\text{start}}}{B-1} \tag{5}$$

where $i = 1, 2, \ldots, B$.

Specifically, for datasets where preprocessing results in noncontiguous bands (removal of bad bands), the central wavelengths are calculated separately for each contiguous segment. The total number of preserved bands, B, is assigned across these segments according to their spectral range.

2) Construction of SRC for Each Domain: Since spectral characteristics differ between domains, we employ the SRC to connect their original spectral spaces to a homogeneous feature space, specifically $S \to \mathcal{X}_h$ and $T \to \mathcal{X}_h$. The process involves several steps as follows.

First, the SRF of a single channel is simulated using a Gaussian function

$$w[i] = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(i - \frac{l-1}{2}\right)^2}{2\sigma^2}\right) \tag{6}$$

where σ is derived from the full width at half maximum (FWHM) as $\sigma = (\text{FWHM}/2.355)$ and l represents the kernel size, with $i = 0, 1, \ldots, l-1$. The FWHM value is set according to the specific FWHM of each dataset, obtainable through published literature, dataset metadata, or by estimating from the spectral range and band number.

Next, expand the SRF to encompass all dimensions ($\lambda_1 \sim \lambda_B$). To determine the convolving location for each dimension, use the central wavelength list μ of the original spectral space and calculate the pairwise distance matrix D

$$D_{ij} = \sqrt{\left(\lambda_i - \mu_j\right)^2}. \tag{7}$$

Then, based on D, we construct a location matrix (L) that determines how the simulated SRF will be applied to each dimension (wavelength)

$$L_{ij} = \begin{cases} 1, & \text{if } D_{ij} = \min\left(D_{i,:}\right) = \min\left(D_{:,j}\right) \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

Following this, the SRC mapping M is derived by convolving L with w column-by-column, followed by normalizing each column of the resulting matrix

$$M = L * w \tag{9}$$

$$M' = \left[\frac{M_{:,1}}{\sum_i M_{i,1}}, \frac{M_{:,2}}{\sum_i M_{i,2}}, \ldots, \frac{M_{:,j}}{\sum_i M_{i,j}}, \ldots\right] \quad \forall j. \tag{10}$$

Following the steps outlined above, we obtain two separate SRC mappings: $M'_{S\to\mathcal{X}_h}$ for the source data and $M'_{T\to\mathcal{X}_h}$ for the target data.

3) Application of SRC for Each Domain: Finally, both the source and target data are transformed into the homogeneous feature space ($\mathcal{X}_h$) by applying their corresponding SRC mappings. Since all pixels within a single domain undergo the same transformation, matrix multiplication offers an efficient way to perform this operation

$$\begin{aligned} \underset{(H\times W)\times C}{X} &\xleftarrow{\text{reshape}} \underset{H\times W\times C}{I} \\ \underset{(H\times W)\times B}{X_{\text{new}}} &= \underset{(H\cdot W)\times C}{X} \times \underset{C\times B}{M'} \\ \underset{H\times W\times B}{I_{\text{new}}} &\xleftarrow{\text{reshape}} \underset{(H\times W)\times B}{X_{\text{new}}}. \end{aligned} \tag{11}$$

TABLE I
ARCHITECTURE OF S2RN

Module	Layer		Filter size†	Padding	Stride	Output size‡
Spectral	conv1		3×1×1, D/2	1,0,0	1,1,1	C×W×H×D/2
	conv2		3×1×1, D	1,0,0	1,1,1	C×W×H×D
	Residual	conv3	1×1×1, D	0,0,0	1,1,1	C×W×H×D
		conv4	1×1×1, D	0,0,0	1,1,1	C×W×H×D
	SE-Net	conv5	1×1×1, D/4	0,0,0	1,1,1	C×W×H×D/4
		conv6	1×1×1, D	0,0,0	1,1,1	C×W×H×D
	conv7		C×1×1, D	0,0,0	1,1,1	W×H×D
Spatial	conv8		3×3, D	1,1	1,1	W×H×D
	conv9		3×3, D	1,1	1,1	W×H×D
Classifier	conv10		3×3, K	1,1	1,1	W×H×K
	GAP					1×1×K

† D is the embedding feature dimension, K is the number of class.
‡ C is the channel size of the input data, W and H are both equal to patch size (P).

B. Spectral–Spatial Residual Network (S2RN) Backbone

The backbone, S2RN, comprises three main components: a spectral module, a spatial module, and a classifier, detailed in Table I and Fig. 4. Moreover, to address challenges in few-shot tasks within the TD, we also introduce the RDD loss function (see Fig. 5).

1) Spectral Module: The spectral module is responsible for extracting and processing spectral information from the input hyperspectral data. First, two 1-D conv layers extract spectral information from the input data. Second, an enhanced residual connection with a squeeze-and-excitation network (SE-Net) [53] is employed to dynamically recalibrate channel-wise responses, highlighting informative spectral features. Third, a 1-D conv layer with a kernel size of $C \times 1 \times 1$ squeezes the channel dimension after spectral feature extraction.

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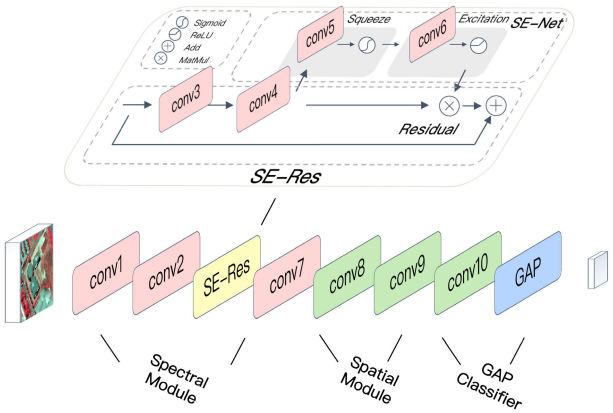


Fig. 4. Illustration of S2RN.

2) Spatial Module: Two 2-D conv layers are utilized to capture spatial patterns among spectral features across adjacent pixels, enabling the module to recognize spatial patterns that signify specific classes within the HSI.

3) Global Average Pooling (GAP) Classifier: A 2-D conv layer with K filters is employed as a fully connected layer, mapping the D-dimensional feature map to K class probabilities. Following this, a GAP layer is applied for spatial aggregation to extract the final classification results.

4) RDD Loss: Based on [54], RDD loss is applied to S2RN to enhance feature discriminability by minimizing intraclass and maximizing interclass distances. Initially, output features from the conv9 layer are extracted for the training batch $\{x, y\}$

$$a = f_{\theta(:\ \text{conv9})}(x). \tag{12}$$

Next, masks for same-class (Λ^+) and different-class (Λ^-) pairs are created

$$\Lambda^+(i,j), \Lambda^-(i,j) = \begin{cases} 1, 0, & \text{if} \quad y_i = y_j \quad \text{and} \quad i \neq j \\ 0, 1, & \text{if} \quad y_i \neq y_j \quad \text{and} \quad i \neq j \\ 0, 0, & \text{otherwise.} \end{cases} \tag{13}$$

Subsequently, pairwise distances (d) using cosine similarity are calculated, applying masks to distinguish between interclass (d^+) and intraclass (d^-) distances

$$d = \left[1 - \frac{a_i \cdot a_j}{\|a_i\| \times \|a_j\|}\right] \quad \forall i, j \tag{14}$$

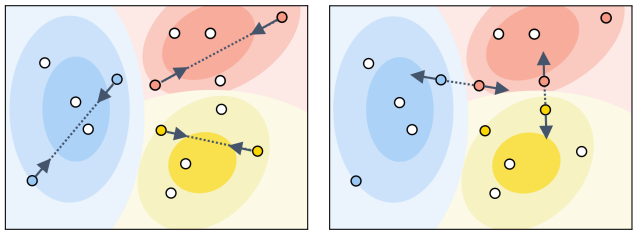
$$d^+ = d \cdot \Lambda^+; \quad d^- = d \cdot \Lambda^-. \tag{15}$$

Then, we apply the logsumexp (LSE) function to find the maximum of d^+ and d^-. The LSE function maximizes d^+ and d^-

$$\text{LSE}(d) = \log\left(\sum_{i=1}^{b} e^{d(i)}\right) \quad \forall i. \tag{16}$$

A scale factor α is applied to ensure numerical precision

$$d^+_{\max} = \text{LSE}\left(\alpha \cdot d^+\right)/\alpha; \quad d^-_{\max} = \text{LSE}\left(\alpha \cdot d^-\right)/\alpha. \tag{17}$$



(a) (b)

Fig. 5. Illustration of RDD loss. (a) Minimize intraclass distances. (b) Maximize interclass distances.

TABLE II
KSN DATA PREPARATION AND USAGE

Data usage	Data preparation				
	Set	Batch	Data		
Pretraining	$\mathcal{L}_S$		x_S, y_S		
Homo-KD	$\mathcal{L}_T$	$B_{\mathcal{U}}$		x^u	$\tilde{x}^u$
Hetero-MT		$B_{\mathcal{L}}$		x, y	$\tilde{x}, y$
Model			f_{θ_p}	f_{θ_h}	f_{θ_e}
Feature space			$\mathcal{X}_h$	$\mathcal{X}_h$	$\mathcal{X}_e$
Domain			SD	TD	TD

Finally, the RDD loss for each batch sample is computed as

$$\ell_{\text{RDD}} = \frac{1}{b}\sum_{i=1}^{b} \frac{d^+_{\max}(i)}{d^+_{\max}(i) + d^-_{\max}(i)}. \tag{18}$$

It is important to note that the RDD loss is deactivated during training in SD, where there are typically more samples compared to the TD. With abundant training data, the focus shifts from maximizing feature discriminability, which RDD loss targets, toward learning generalizable spectral features suitable for broader applications.

C. Knowledge Swap Net

KSN operates in four stages. Initially, homogeneous spectral knowledge is transferred from SD to TD via pretraining. Next, the homogeneous spectral knowledge is distilled into a heterogeneous model. Then, discriminative knowledge from the heterogeneous model is fed back to the homogeneous model. Finally, the models are fused to make the prediction.

Before delving into KSN in detail, the setup of the training data is presented as follows (see Table II).

For the SD data, we first employ SRC to map it into homogeneous feature space $\mathcal{X}_h$. The obtained labeled sample set, $\mathcal{L}_S = \{(x_S, y_S)\}$, are used for training the SD pretrained model f_{θ_p}. To process the TD data, SRC and PCA are employed to project it into the homogeneous ($\mathcal{X}_h$) and heterogeneous ($\mathcal{X}_e$) feature spaces, represented by x_T and $\tilde{x}_T$, respectively. For clarity, the T subscript is omitted for x_T and $\tilde{x}_T$ hereafter. From the limited labeled sample set in TD $\mathcal{L}_T = \{(x, \tilde{x}, y)\}$, we create two equally sized minibatches: $B_{\mathcal{L}} = \{(x, \tilde{x}, y)\}$, retaining labels, and $B_{\mathcal{U}} = \{(x^u, \tilde{x}^u)\}$, where labels are omitted to form an unlabeled batch. The unlabeled minibatch, $B_{\mathcal{U}}$, supports heterogeneous KD, while the labeled minibatch, $B_{\mathcal{L}}$, is used for homogeneous ML. Data from the homogeneous space (plain x) is fed into the homogeneous model (f_{θ_h}), and data from the heterogeneous space (denoted as $\tilde{x}$) goes to the heterogeneous model (f_{θ_e}).

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1) Homogeneous Pretraining: Starting with the homogenized SD training data $\mathcal{L}_S$, the goal is to optimize the parameters of the pretrained model f_{θ_p} to minimize cross-entropy loss

$$\ell_p \leftarrow \frac{1}{|\mathcal{L}_S|}\sum \ell_{\mathrm{CE}}\left(f_{\theta_p}, x_S, y_S\right) \tag{19}$$

$$\theta_p \leftarrow \theta_p - \eta \cdot \nabla_{\theta_p}\ell_p \tag{20}$$

where θ is initialized following the normal distribution $\mathcal{N}(0, 0.001)$. Following this, the homogeneous model f_{θ_h} is constructed and initialized

$$\theta_h^{(0)}(:\ \mathrm{conv9}) = \theta_p(:\ \mathrm{conv9}) \tag{21}$$

$$\theta_h^{(0)}(\mathrm{conv10, GAP}) \sim \mathcal{N}(0, 0.001). \tag{22}$$

Specifically, the GAP classifier in f_{θ_h} is newly constructed for the target differing tasks and initialized following the normal distribution.

The above procedure ensures that f_{θ_h} retains the learned features from SD, with its independently initialized GAP Classifier enabling tailored classification for the TD.

2) Homogeneous KD: In this stage, rather than immediately fine-tuning the homogeneous model f_{θ_h}, we first compute its loss value on the labeled training batch $B_{\mathcal{L}}$

$$\ell_h \leftarrow \frac{1}{|B_{\mathcal{L}}|}\sum \ell_{\mathrm{CE+RDD}}\left(f_{\theta_h}, x, y\right). \tag{23}$$

Subsequently, f_{θ_h} generates pseudo-labels for the unlabeled training batch $B_{\mathcal{U}}$, defined as $B_{\mathcal{U}} = \{(x^u, \tilde{x}^u, \hat{y}_u)\}$

$$\hat{y}^u = f_{\theta_h}\left(x^u\right). \tag{24}$$

With the pseudo-labels, the heterogeneous model f_{θ_e} is then trained to optimize KL divergence (1), encouraging it to mimic f_{θ_h}, forming the learning process in $\mathcal{X}_h$. To modulate the confidence of the pseudo-labels, $\hat{y}^u$ are adjusted using (2). The processes can be mathematically represented as follows:

$$\hat{y}_\tau^u = \left[\sigma_\tau\left(\hat{y}_c^u\right)\right] \quad \forall c \tag{25}$$

$$\ell_e \leftarrow \frac{1}{|B_{\mathcal{U}}|}\sum \ell_{\mathrm{KL+RDD}}\left(f_{\theta_e}\left(\tilde{x}^u\right)\middle\|\hat{y}_\tau^u\right) \tag{26}$$

$$\theta_e \leftarrow \theta_e - \eta \cdot \nabla_{\theta_e}\ell_e \tag{27}$$

where θ_e is initialized randomly with $\mathcal{N}(0, 0.001)$, ensuring f_{θ_e} to independently adapt to $\mathcal{X}_e$.

3) Heterogeneous Meta-Learning: The quality of the pseudo-labels directly affects the performance of f_{θ_e}. If f_{θ_e} performs well, it implies f_{θ_h} is generating high-quality pseudo-labels; if not, θ_h might need adjustments. To enable learning such feedback from f_{θ_e} by f_{θ_h}, we propose a Hetero-ML approach inspired by meta pseudo labels (MPL) [55] for guiding the update of θ_h. The approach leveraging the test performance of f_{θ_e} to evaluate the quality of the pseudo-labels

$$\ell_{\mathrm{meta}}^{(\varepsilon)} = \frac{1}{|B_{\mathcal{L}}|}\sum \ell_{\mathrm{CE}}\left(f_{\theta_e^{(\varepsilon)}}(\tilde{x}), y\right) \tag{28}$$

where the held-out set $\tilde{x}, y$ has not been seen by f_{θ_e} during the εth epoch.

To monitor the performance variation of f_{θ_e}, we need to compare its losses between different meta-stages. Unlike MPL, which optimizes θ_h based on the test loss of f_{θ_e} before and after a single update, our method assesses the performance of f_{θ_e} over consecutive epochs solely after each update

$$\Delta\ell_{\mathrm{meta}}^{(\varepsilon)} = \ell_{\mathrm{meta}}^{(\varepsilon)} - \ell_{\mathrm{meta}}^{(\varepsilon-1)}. \tag{29}$$

Thereby, not only reducing the computational load for testing but also allowing for a more thorough analysis of long-term performance trends and stability.

Finally, the testing performance variation of f_{θ_e}, indicated by $\Delta\ell_{\mathrm{meta}}$, guides the update of the homogeneous model

$$\theta_h^{(\varepsilon)} \leftarrow \theta_h^{(\varepsilon-1)} - \eta \cdot r \cdot \nabla_{\theta_h^{(\varepsilon-1)}}\ell_h^{(\varepsilon-1)} \tag{30}$$

where $r = 1 - \Delta\ell_{\mathrm{meta}}$ serves as a feedback factor, with η being the learning rate.

A concern with our modification is the held-out set variability between epochs, which might affect the testing performance of f_{θ_e} and not truly reflect the performance of f_{θ_h}. To address this, particularly in the few-shot scenario of TD, KSN utilizes the batch including all training samples, randomly selecting half for testing f_{θ_e} in each epoch. Therefore, the frequent reappearance of samples lessens the impact of variability.

4) Prediction With Additive Fusion: Finally, we derive the final prediction by combining outputs from both models to ensure performance stability, as disagreements may still arise on unseen test samples

$$\hat{y} = f_{\theta_h^*}(x) + f_{\theta_e^*}\left(\tilde{x}\right). \tag{31}$$

Up to this point, we have outlined the core elements of the KSN, detailed in Algorithm 1.

Algorithm 1 Pseudo Code of the Proposed KSN

Require: $\mathcal{L}_S, \mathcal{L}_T, E_S, E_T, \eta_S, \eta_T, \theta_p^{(0)}, \theta_e^{(0)}, \ell_{CE}, \ell_{KL}, \ell_{RDD}, \sigma_\tau$

I. Homogeneous Pertraining:
for *step* $\varepsilon \in \{1 \ldots E_S\}$ **do**
  $\{x_s, y_s\} \leftarrow RandSample(\mathcal{L}_S)$
  $\ell_p^{(\varepsilon-1)} = \frac{1}{|\mathcal{L}_S|}\sum \ell_{CE}(f_{\theta_p^{(\varepsilon-1)}}(x_s), y_s)$
  $\theta_p^{(\varepsilon)} \leftarrow \theta_p^{(\varepsilon-1)} - \eta_S \cdot \nabla_{\theta_p^{(\varepsilon-1)}}\ell_p^{(\varepsilon-1)}$ ▷ *Update* θ_p
$\theta_h^{(0)} \leftarrow \theta_p^*$ ▷ *Weight transfer*
for *step* $\varepsilon \in \{1 \ldots E_T\}$ **do**
  $\{B_{\mathcal{L}}, B_{\mathcal{U}}\} \leftarrow RandBisect(\mathcal{L}_T)$
  /* $B_{\mathcal{L}} = \{(x, \tilde{x}, y)\}$, $B_{\mathcal{U}} = \{(x^u, \tilde{x}^u)\}$ */
  II Homogeneous Knowledge Distillation:
  $\ell_h^{(\varepsilon-1)} = \frac{1}{|B_{\mathcal{L}}|}\sum \ell_{CE+RDD}(f_{\theta_h^{(\varepsilon-1)}}, x, y)$
  $\hat{y}^u = f_{\theta_h^{(\varepsilon-1)}}(x^u)$ ▷ $B_{\mathcal{U}} = \{(x^u, \tilde{x}^u, \hat{y}_u)\}$
  $\ell_e^{(\varepsilon-1)} = \frac{1}{|B_{\mathcal{U}}|}\sum \ell_{KL+RDD}(f_{\theta_e^{(\varepsilon-1)}}, \tilde{x}^u, \sigma_\tau(\hat{y}^u))$
  $\theta_e^{(\varepsilon)} \leftarrow \theta_e^{(\varepsilon-1)} - \eta_T \cdot \nabla_{\theta_e^{(\varepsilon-1)}}\ell_e^{(\varepsilon-1)}$ ▷ *Update* θ_e
  III Heterogeneous Meta-learning:
  $\ell_{meta}^{(\varepsilon)} = \frac{1}{|B_{\mathcal{L}}|}\sum \ell_{CE+RDD}(f_{\theta_e^{(\varepsilon)}}, \tilde{x}, y)$
  $\Delta\ell_{meta}^{(\varepsilon)} = \ell_{meta}^{(\varepsilon)} - \ell_{meta}^{(\varepsilon-1)}$ ▷ $\Delta\ell_{meta}^{(1)} = 0$
  $r = 1 - \Delta\ell_{meta}^{(\varepsilon)}$
  $\theta_h^{(\varepsilon)} \leftarrow \theta_h^{(\varepsilon-1)} - \eta_T \cdot r \cdot \nabla_{\theta_h^{(\varepsilon-1)}}\ell_h^{(\varepsilon-1)}$ ▷ *Update* θ_h
IV. Prediction with Additive Fusion:
$\hat{y} = f_{\theta_h^*}(x) + f_{\theta_e^*}(\tilde{x})$ ▷ *Improve stability*

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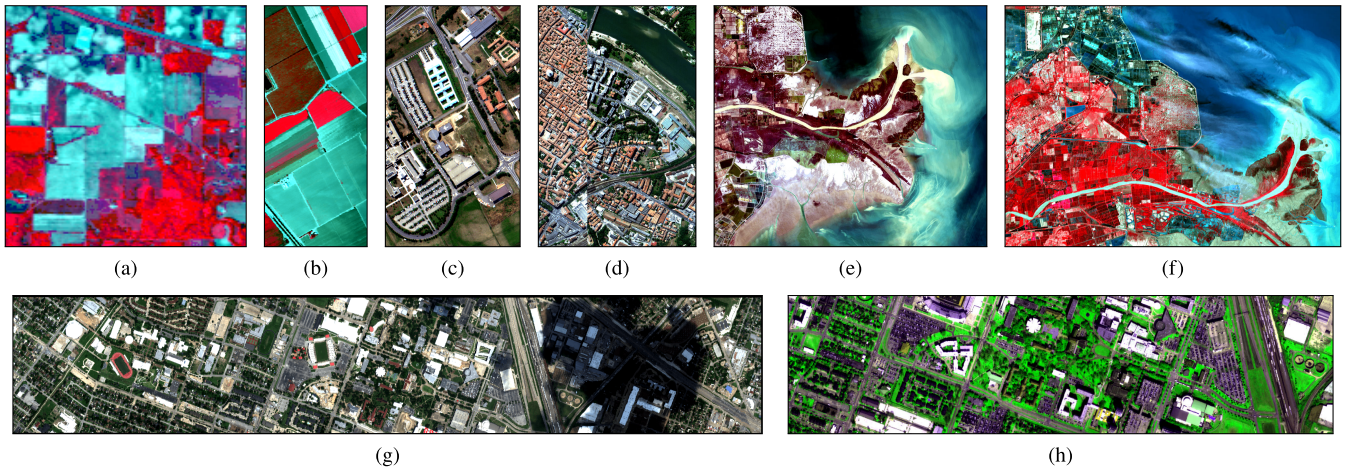


Fig. 6. False-color composite images of (a) Indian Pines (IP), (b) Salinas, (c) Pavia University (PU), (d) Pavia City, (e) GF-5 Yellow River Delta (GF), (f) ZY-1 02D Yellow River Delta, (g) Houston University (HU) 2013, and (h) HU 2018.

TABLE III
SPECIFICATIONS FOR DIFFERENT HYPERSPECTRAL DATASETS

Dataset	Sensor	Range (μm)	FWHM (μm)	# Bands	# Reduced bands	GSD (m)	Image size	# Class	# Labeled
Indian Pines **(IP)**	AVIRIS	400∼2500	10/15 (VN/SWIR)	224	200	20 m	145×145	16	10249
Salinas **(SA)**					204	3.7 m	512×217	16	53785
Pavia University **(PU)**	ROSIS	430∼860	4	115	103	1.3 m	610×340	9	42776
Pavia City **(PC)**					102	1.3 m	1096×715	9	148152
Houston University 2013 **(HU)**	CASI-1500	380∼1050	4.8	288	144	2.5 m	349×1905	15	15029
Houston University 2018 **(HU18)**			14.4		48	1 m	1202×4172	20	504712
GF-5 Yellow River Delta **(GF)**	GF-5 AHSI	390∼2500	5/10 (VN/SWIR)	330	285	30 m	1185×1324	23	6471
ZY-1 02D Yellow River Delta **(ZY)**	ZY-1 02D AHSI	400∼2500	10/20 (VN/SWIR)	166	119	30 m	1147×1600	21	9825

TABLE IV
NUMBER OF SAMPLES PER CLASS FOR FOUR TARGET DATASETS

PU			IP			HU			GF		
Class	Name	#	Class	Name	#	Class	Name	#	Class	Name	#
1	Asphalt	6631	1	Alfalfa	46	1	Healthy grass	1251	1	Aquaculture	393
2	Meadows	18649	2	Corn-notill	1428	2	Stressed grass	1254	2	Seep sea	796
3	Gravel	2099	3	Corn-mintill	830	3	Synthetic grass	697	3	Soybean	110
4	Trees	3064	4	Corn	237	4	Trees	1244	4	Rice	190
5	Painted metal sheets	1345	5	Grass-pasture	483	5	Soil	1242	5	Building	83
6	Bare soil	5029	6	Grass-trees	730	6	Water	325	6	Maize	96
7	Bitumen	1330	7	Grass-pasture-mowed	28	7	Residential	1268	7	Broomcorn	95
8	Self-blocking bricks	3682	8	Hay-windrowed	478	8	Commercial	1244	8	Locust	211
9	Shadows	947	9	Oats	20	9	Road	1252	9	Spartina	200
			10	Soybean-notill	972	10	Highway	1227	10	Shallow sea	936
			11	Soybean-mintill	2455	11	Railway	1235	11	Mud flat	553
			12	Soybean-clean	593	12	Parking lot 1	1233	12	River	469
			13	Wheat	205	13	Parking lot 2	469	13	Suaeda salsa	361
			14	Woods	1265	14	Tennis court	428	14	Reed	240
			15	Buildings-Grass-Trees-Drives	386	15	Running track	660	15	Salt marsh	595
			16	Stone-Steel-Towers	93				16	Intertidal saltwater	454
									17	Tamarix	133
									18	Pond	377
									19	Flood plain	68
									20	Freshwater herbaceous marsh	72
									21	Aquatic vegetation	39

IV. EXPERIMENTAL RESULTS

A. Experiment Data

1) Hyperspectral Datasets: False-color composite images of the hyperspectral datasets used in experiments are displayed in Fig. 6. Table III details the specifications of these datasets. The TD datasets include PU, IP, HU, and GF (detailed in Table IV), and each of them is paired with a corresponding SD dataset (PC, SA, HU18, and ZY, respectively). These datasets were chosen for their spectral diversity and land cover representativeness.

2) Sample Selection: For the source datasets in our study, we employed a pretraining model using 500 samples per class, or half that number for classes with fewer samples. The source data quantity was determined based on subsequent parameter sensitivity analyses. In the TDs, unless otherwise specified, we utilized a five-shot classification approach as a baseline, which uses only five labeled samples per class from each target

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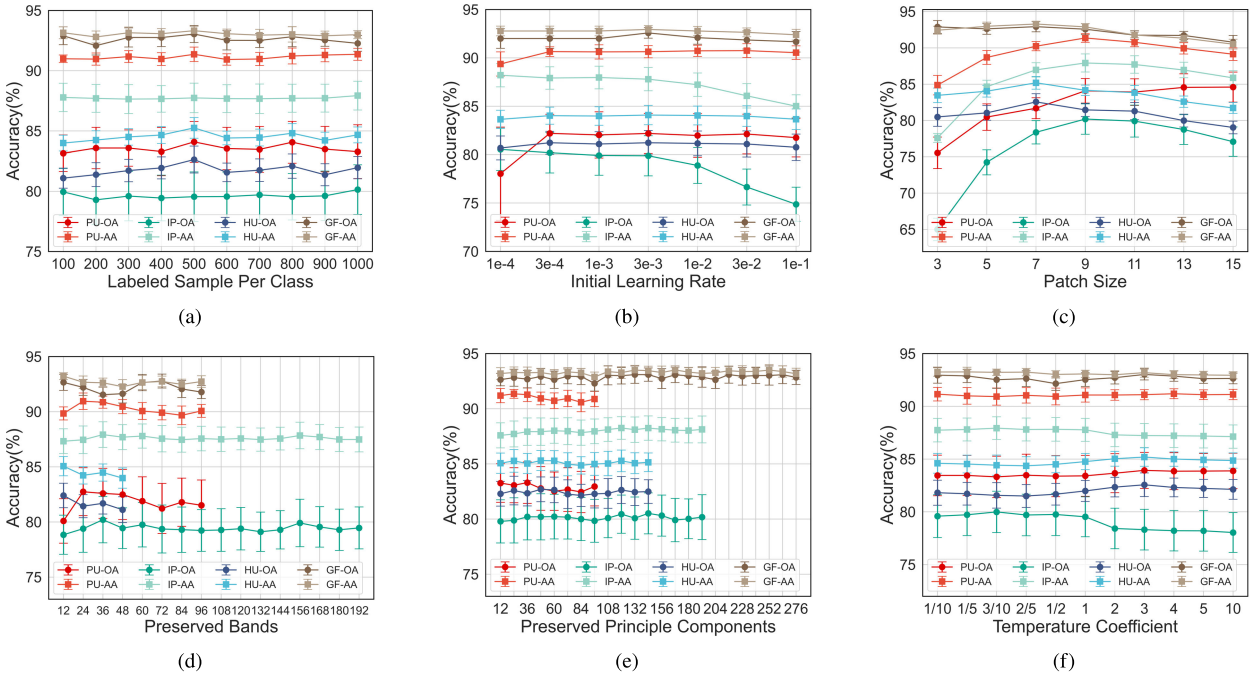


Fig. 7. Parameter sensitivity analysis of KSN. (a) Proportions of pretraining data. (b) Initial learning rates. (c) Patch sizes. (d) Numbers of preserved bands. (e) Numbers of preserved principal components. (f) Temperatures coefficient.

TABLE V
OPTIMIZATION CONFIGURATIONS FOR PRETRAINING AND KNOWLEDGE SWAP PHASES

Hyperparameter	Pretraining	Knowledge swap
Batch size (b)	64	# training samples
Epoch (E)	100	400
Initial learning rate (η_0)	1e-3	3e-3
Learning rate schedule	fixed	cosine annealing
Momentum of SGD	0.9	0.9
Weight decay of SGD	5e-4	5e-4

dataset. The remaining labeled samples were then used as test samples to evaluate the classification performance.

B. Experimental Setting

1) Optimization: The hyperparameters utilized for optimization are detailed in Table V. For both the pretraining and knowledge swap phase, stochastic gradient descent (SGD) is employed as the optimizer with a momentum of 0.9 and weight decay of $5e-4$. During the pretraining phase, the pretrained model is trained at an initial learning rate of 0.001 across 100 epochs, with a batch size of 64. In the knowledge swap phase, which integrates both homogeneous and heterogeneous models, the learning rate follows a cosine annealing strategy. It starts at $3e-3$, and the maximum number of iterations is set equal to the total epochs, which are set at 400.

2) Compared Methods: Our method was compared with several state-of-the-art FSL and cross-domain classification methods from the hyperspectral classification community. For FSL, this includes S3Net [52] and SPFormer [58]. The cross-domain classification methods we considered are DCFSL [38], CMFSL [48], HFSL [56], Gia-CFSL [57], FSCF-SSL [18], ADAFSL [36], and Lee et al.'s cross-domain pretraining method [47]. Detailed information about these methods is compiled in Table VI, and the SD configurations used in the cross-domain methods are outlined in Table VII.

3) Running Platforms: Experiments were conducted using an Intel Core i7-13700KF CPU, GeForce RTX 3080 (10 GB), and 64 GB RAM. The software environment included Python 3.8, PyTorch 1.13.1, and cuDNN 11.7.1.

4) Evaluation and Metrics: For a fair comparison, each method underwent ten independent trials using random seeds, with results presented as the average of these trials. Performance was evaluated using metrics such as class accuracy (CA), overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ).

C. Parameter Sensitive Analysis

In this section, we examine the performance variation of KSN by analyzing key hyperparameters in a five-shot setting. Sensitivity analysis standardizes most hyperparameters and optimizes dataset-specific ones like principal components.

1) Source Training Sample Size (m): Exploring the impact of SD sample size (m), we found that 500 samples per class yielded optimal accuracy for the PU and HU datasets [see Fig. 7(a)]. The IP and GF datasets exhibited less sensitivity to variations in m. Consequently, we adopted a consistent SD size of 500 samples per class for subsequent experiments.

2) Initial Learning Rate (η_0): We evaluated initial learning rates (η_0) ranging from $1e-4$ to $1e-1$. While the HU and GF datasets showed relative insensitivity, the PU dataset benefited from higher learning rates (above $3e-4$), and the IP dataset performed better with lower rates [see Fig. 7(b)]. To balance performance and stability across datasets, we set η_0 to $3e-3$ for subsequent experiments.

3) Patch Size (P): Patch size (P) significantly influenced classification accuracy [see Fig. 7(c)]. The PU dataset exhib-

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TABLE VI
COMPARISON OF COMPARED METHODS

Method	Cross-Domain	FSL	Pretrain	Architecture	Year	Code
Lee et al. [47]	✓		✓	Fine-tuning	2022	
DCFSL [38]	✓	✓		Proto. Net	2021	Link
CMFSL [48]	✓	✓		Proto. Net	2022	Link
HFSL [56]	✓	✓	✓	Proto. Net	2022	Link
Gia-CFSL [57]	✓	✓		Proto. Net	2023	Link
FSCF-SSL [18]	✓	✓	✓	Proto. Net	2023	Link
ADAFSL [36]	✓	✓		Proto. Net	2023	Link
S3Net [52]		✓		Siamese Net	2022	Link
SPFormer [58]		✓	✓	Transformer	2023	Link

TABLE VII
SD CHOICES FOR CROSS-DOMAIN METHODS. NEWLY INCLUDED DATASETS: CHIKUSEI, KENNEDY SPACE CENTER (KSC), BOTSWANA (BW), AND MINIIMAGENET. "C.K.B." DENOTES THE COMBINATION OF CHIKUSEI, KSC, AND BW DATASETS

Methods	Target Dataset			
	PU	IP	HU	GF
Lee et al	PC	Salinas	HU18	ZY
DCFSL	CKS	CKS	CKS	CKS
CMFSL	C.K.B.+HU	C.K.B.+HU	C.K.B.	C.K.B.+HU
HFSL	MiniImageNet	MiniImageNet	MiniImageNet	MiniImageNet
Gia-CFSL	CKS	CKS	CKS	CKS
FSCF-SSL	MiniImageNet	MiniImageNet	MiniImageNet	MiniImageNet
ADAFSL	CKS	CKS	CKS	CKS
KSN(ours)	PC	Salinas	HU18	ZY

ited improved performance with larger patch sizes (up to $P = 9$), likely due to the presence of extensive contiguous regions. However, further increasing the patch size led to a decrease in AA due to the potential blurring of boundaries for smaller land features. The IP dataset exhibited a similar trend in both OA and AA and found optimal $P = 9$. Conversely, the GF and HU datasets favored smaller patch sizes ($P = 7$) due to the predominance of smaller land cover types.

4) Preserved Band Number of Spectral Resampling (B): Fig. 7(d) illustrates the impact of the number of preserved bands in spectral resampling. For the PU dataset, the optimal number of preserved bands (B) is 24, while IP benefits from a higher B at 36, enhancing model efficiency. HU and GF datasets achieve ideal performance with B set to 12.

5) Number of Preserved Principle Components (C): The number of preserved principal components (C) showed minimal impact on performance across datasets [see Fig. 7(e)]. We set C to 36 for PU and HU, and 48 for IP and GF to balance efficiency and spectral information preservation.

6) Temperature Coefficient (τ): We tested 11 values of temperature coefficient (τ) to assess its impact on KSN [see Fig. 7(f)]. Optimal τ varied: we found $\tau = 0.3$ to be beneficial for the IP dataset, while $\tau = 3$ improved performance on PU, HU, and GF, indicating different self-supervised learning needs.

7) Embedding Feature Dimensions (D): We investigated embedding dimensions of 36, 64, and 128 for both homogeneous (D_h) and heterogeneous (D_e) models (see Fig. 8). Larger dimensions generally improved performance, but we chose $D_h = D_e = 64$ to balance performance and computational efficiency.

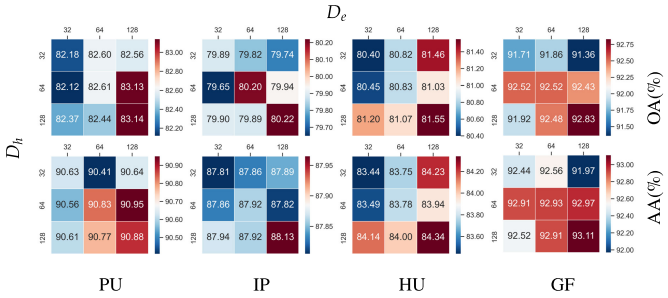


Fig. 8. Parameter sensitive analysis of the embedding feature dimensions.

D. Comparison of Classification Performance

Tables VIII–XI present classification accuracy comparisons of various methods. Across four target datasets, KSN stood out as the top performer in OA for IP and HU, and was among the top for PU and GF. In addition, it led in AA for all datasets and secured the best κ for IP, HU, and GF, with a close second on PU.

1) On the PU dataset, while HFSL and FSCF-SSL achieved higher OA of 86.71% and 88.04%, respectively, KSN demonstrated AA of 91.36%, indicating that our method provides better class balance compared to the others. A closer examination of category-wise accuracies reveals that the exceptional performance of HFSL and FSCF-SSL is primarily driven by their accurate classification of the large-area Meadow category (Index 2), likely due to their larger patch size of 33 × 33 pixels. While most compared methods surpassed an 80% OA, only KSN, HFSL, and FSCF-SSL exceeded 90% in AA and 0.8 in κ.
2) KSN demonstrated superior accuracy on the IP dataset, outperforming the runner-up SPFormer in OA, AA, and κ. KSN also recorded the highest CA in 9 of 16 categories and ranked second in 3. In this task, episodic training methods underperformed, particularly in six specific classes, including Corn-notill, Corn-mintill, Corn, Soybean-notill, Soybean-mintill, Soybean-clean (indices 2, 3, 4, 10, 11, and 12), indicating limitations in fine-grained classification.
3) Regarding the HU dataset, KSN maintained superior performance in terms of OA, AA, and κ, surpassing other methods by 3.83%–7.97%, 4.35%–9.24%, and 0.04-0.09, respectively. KSN achieved an OA of 82.56%, AA of 85.19%, and κ of 0.81, while the best-performing competitor (Lee et al.'s method) reached an OA of 78.73%, AA of 79.79%, and $\kappa$of 0.77. Other methods, including CMFSL, S3Net, and SPFormer, also exceeded 77% OA but demonstrated variations in category-wise accuracy. While these methods excelled in certain categories like healthy grass (index 1) and trees (index 4), KSN's consistent performance across all classes contributed to its overall superior accuracy.
4) On the GF dataset, the performance of KSN matched that of Lee et al., but with a higher AA. The lower AA of Lee et al. resulted from poor performance in the flood plain (index 19). Among the compared methods, FSL approaches like S3Net and SPFormer underperformed,

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TABLE VIII

CLASSIFICATION RESULTS ON THE PU DATASET, WITH FIVE SAMPLES PER CLASS

Class	Lee et al. [47]	DCFSL [38]	CMFSL [48]	HFSL [56]	Gia-CFSL [57]	FSCF-SSL [18]	ADAFSL [36]	S3Net [52]	SPFormer [58]	KSN (Ours)
1	88.73±6.61	73.78±9.56	75.33±9.93	75.28±11.87	77.69±9.77	74.77±9.20	82.00±6.66	88.53±5.30	83.95±8.10	**90.43±4.66**
2	82.94±6.24	81.72±9.93	78.76±12.91	87.35±7.10	83.12±9.46	**88.78±4.44**	77.36±10.51	76.54±7.63	69.65±9.55	74.06±7.59
3	76.59±8.31	68.11±7.88	75.71±14.19	**90.43±5.98**	64.06±8.81	85.68±10.68	74.16±12.83	81.06±12.43	86.52±7.87	89.08±5.87
4	78.76±14.70	**94.90±1.28**	92.61±3.28	92.63±4.93	92.88±4.30	94.13±4.82	88.20±6.43	82.30±8.63	89.43±8.98	91.72±3.68
5	98.17±2.28	99.60±0.90	99.51±0.84	99.41±1.10	99.51±0.51	99.90±0.13	98.54±1.54	99.50±0.96	**99.96±0.11**	99.37±1.25
6	67.81±11.53	72.44±14.03	86.30±7.42	86.46±8.15	73.16±8.71	89.93±7.15	71.72±11.98	89.89±7.22	**92.83±6.62**	92.71±4.91
7	82.46±9.59	80.41±8.70	86.81±7.76	94.59±4.87	82.08±7.52	93.36±7.78	90.10±6.32	99.22±0.86	**99.60±0.91**	99.34±1.10
8	77.55±8.40	63.01±11.67	71.65±11.93	87.11±16.96	68.46±13.85	**92.62±11.08**	64.22±18.69	81.59±15.51	81.42±12.45	88.01±7.14
9	98.22±0.92	98.52±2.24	97.02±4.07	97.45±7.02	96.73±3.33	**99.92±0.13**	97.60±4.16	84.77±8.05	95.91±4.41	97.48±2.93
OA(%)	81.00±5.14	78.96±5.71	80.65±5.60	86.71±2.37	80.39±5.15	**88.04±3.45**	78.41±4.06	82.64±3.81	80.31±3.65	84.09±3.38
AA(%)	83.47±3.62	81.39±3.19	84.85±1.94	90.08±3.42	81.97±2.57	91.01±2.72	82.65±2.31	87.04±2.70	88.81±1.61	**91.36±1.20**
κ	0.76±0.06	0.73±0.07	0.75±0.07	0.83±0.03	0.75±0.06	**0.84±0.05**	0.72±0.05	0.78±0.04	0.76±0.04	0.80±0.04

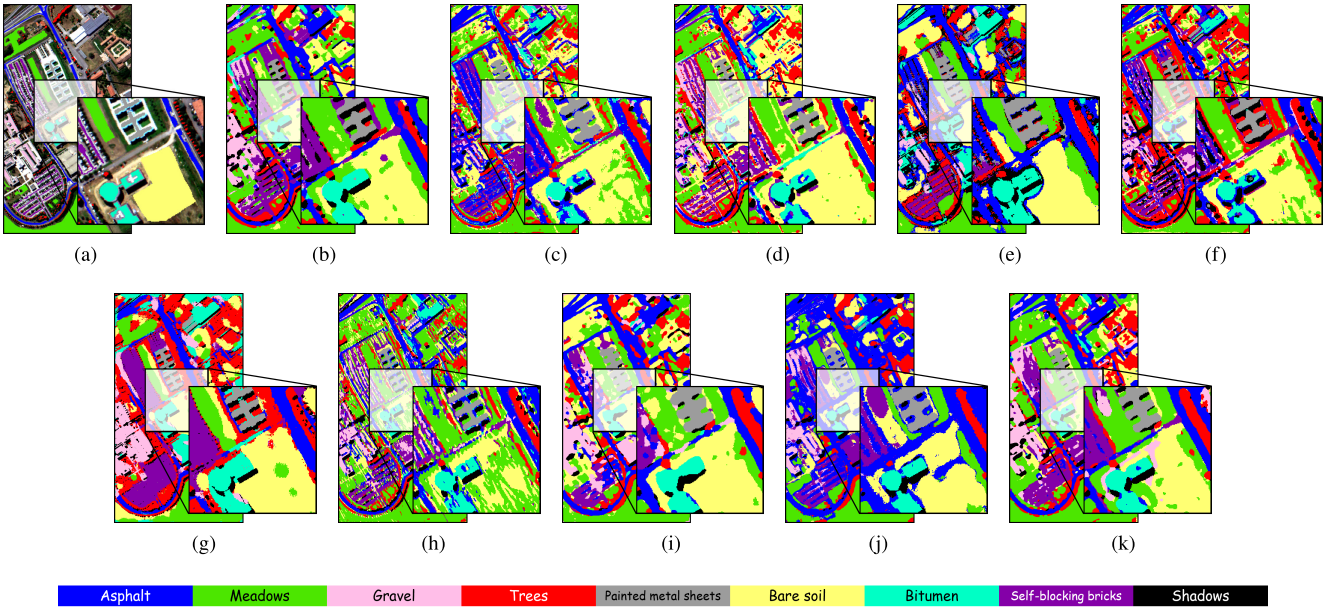


Fig. 9. Classification maps of PU with different methods: (a) ground truth, (b) Lee et al. (OA = 89.39%), (c) DCFSL (OA = 81.40%), (d) CMFSL (OA = 92.13%), (e) HFSL (OA = 78.70%), (f) Gia-CFSL (OA = 90.85%), (g) FSCF-SSL (OA = 87.45%), (h) ADAFSL (OA = 82.03%), (i) S3Net (OA = 86.20%), (j) SPFormer (OA = 82.46%), and (k) KSN (OA = 91.17%).

TABLE IX

CLASSIFICATION RESULTS ON THE IP DATASET, WITH FIVE SAMPLES PER CLASS

Class	Lee et al. [47]	DCFSL [38]	CMFSL [48]	HFSL [56]	Gia-CFSL [57]	FSCF-SSL [18]	ADAFSL [36]	S3Net [52]	SPFormer [58]	KSN (Ours)
1	82.83±16.99	94.88±4.69	95.36±3.69	97.80±2.98	91.22±6.83	98.29±3.10	96.83±3.28	**99.76±0.73**	99.02±1.62	**99.76±0.73**
2	54.99±11.69	49.06±9.43	51.10±10.27	55.74±8.41	52.67±11.59	59.46±5.84	58.30±13.89	67.88±7.72	65.19±11.47	**72.39±5.55**
3	61.31±10.21	45.18±12.47	50.29±13.39	62.70±13.91	44.32±12.15	61.07±16.71	49.33±11.35	58.29±13.64	62.02±18.00	**64.19±16.22**
4	62.20±15.32	78.06±7.18	87.20±9.86	88.84±14.10	73.06±13.24	89.53±11.37	80.39±8.58	96.72±5.09	**97.16±3.94**	96.47±5.60
5	**80.91±9.86**	70.17±6.87	75.56±6.37	72.74±6.37	73.56±7.47	71.36±6.63	74.66±7.77	76.80±8.00	78.72±8.95	77.66±9.36
6	94.98±2.99	89.27±3.01	93.60±4.63	83.57±10.05	83.08±5.94	85.74±7.01	91.21±6.02	93.53±4.35	**95.92±2.44**	95.50±2.52
7	51.90±15.90	**100.00±0.00**	98.70±3.91	**100.00±0.00**	98.70±3.91	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	97.67±2.33	86.07±9.86	89.11±9.15	98.50±2.50	89.09±6.37	98.61±2.76	84.69±10.32	99.30±1.77	99.89±0.22	**99.94±0.19**
9	40.69±12.25	98.00±4.27	100.00±0.00	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.00±6.00	**100.00±0.00**
10	59.73±9.51	63.65±12.25	64.74±10.15	65.78±15.40	63.28±15.18	67.92±14.84	71.70±9.53	75.67±9.38	77.08±10.97	**78.52±9.04**
11	62.05±10.86	56.20±12.34	48.31±14.23	60.05±21.13	53.69±12.46	61.13±17.53	58.14±12.09	68.48±11.70	70.74±10.81	**72.98±11.76**
12	57.92±9.93	40.34±12.81	48.08±11.24	69.00±8.73	42.24±11.92	70.90±10.58	61.00±19.02	67.07±7.13	74.42±13.52	**74.85±10.97**
13	94.43±6.43	98.65±1.79	99.45±0.76	98.60±3.56	96.20±7.26	**99.95±0.15**	98.95±0.79	99.05±2.05	99.45±0.52	99.45±1.35
14	89.16±4.74	86.70±7.52	82.98±7.45	89.42±7.49	80.15±9.98	92.11±4.02	84.05±8.11	93.37±3.95	**94.28±3.49**	92.70±5.80
15	66.66±9.49	74.51±6.57	84.86±7.60	**95.77±6.60**	65.01±5.56	93.36±8.68	84.52±7.48	87.09±8.98	87.77±8.43	82.70±7.34
16	91.18±5.55	98.18±2.97	98.86±2.69	98.86±2.03	98.64±1.96	97.95±3.55	97.27±4.41	98.98±1.19	**100.00±0.00**	99.66±0.52
OA(%)	69.42±4.90	64.98±3.29	65.19±2.81	71.81±4.23	63.40±4.92	73.15±4.46	69.43±4.26	77.32±3.64	78.80±4.55	**80.20±4.19**
AA(%)	71.79±3.73	76.81±2.18	79.26±1.46	83.59±1.50	75.31±3.33	84.21±2.33	80.69±2.81	86.37±2.26	87.48±2.54	**87.92±2.32**
κ	0.66±0.05	0.60±0.04	0.61±0.03	0.68±0.04	0.59±0.06	0.70±0.05	0.66±0.05	0.75±0.04	0.76±0.05	**0.78±0.05**

while cross-domain methods can easily surpassed 90% in OA, highlighting the need for this dataset for strong generalizability. The findings underscore the effectiveness of cross-domain pretraining approaches, as utilized in both KSN and Lee et al.'s method.

Figs. 9–12 display classification maps across four TDs, with KSN demonstrating superior classification accuracy, edge definition, and detail rendering. Overall, KSN reduces misclassifications and noise more effectively than other methods, excelling in the PU, IP, and HU datasets and avoiding

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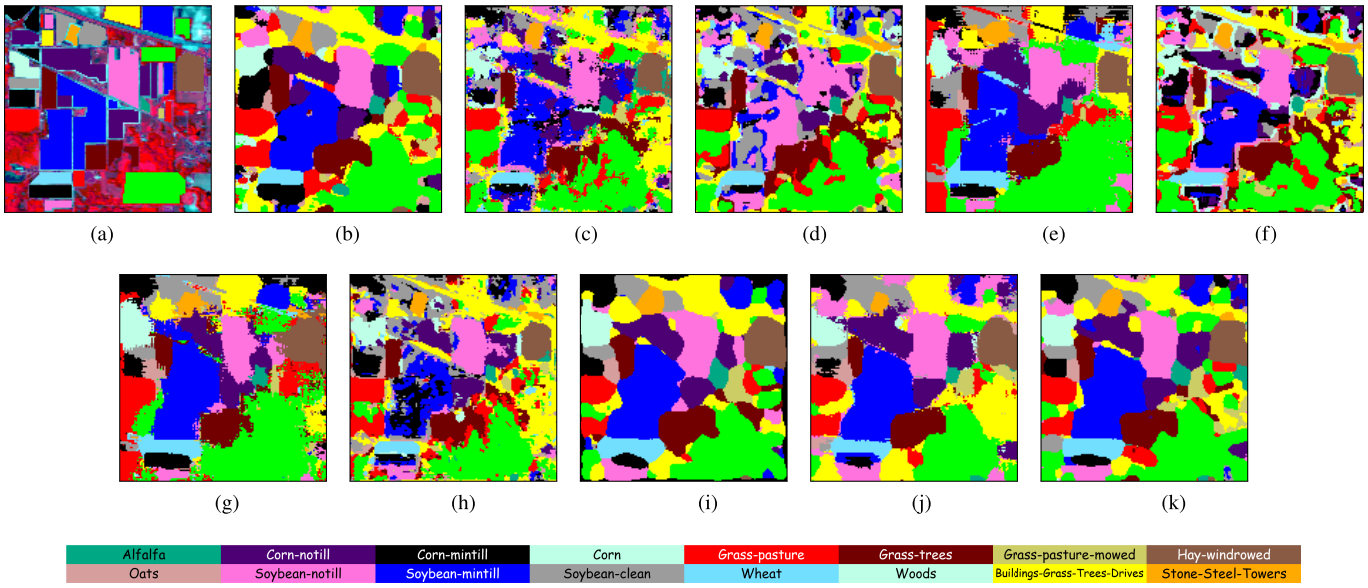


Fig. 10. Classification maps of IP with different methods: (a) ground truth, (b) Lee et al. (OA = 84.25%), (c) DCFSL (OA = 66.29%), (d) CMFSL (OA = 60.59%), (e) HFSL (OA = 78.71%), (f) Gia-CFSL (OA = 64.82%), (g) FSCF-SSL (OA = 81.73%), (h) ADAFSL (OA = 68.21%), (i) S3Net (OA = 83.59%), (j) SPFormer (OA = 85.49%), and (k) KSN (OA = 85.67%).

TABLE X
CLASSIFICATION RESULTS ON THE HU 2013 DATASET, WITH FIVE SAMPLES PER CLASS

Class	Lee et al. [47]	DCFSL [38]	CMFSL [48]	HFSL [56]	Gia-CFSL [57]	FSCF-SSL [18]	ADAFSL [36]	S3Net [52]	SPFormer [58]	KSN (Ours)
1	90.69±3.22	89.31±4.79	90.69±6.13	74.80±11.52	89.88±5.08	78.28±11.67	**91.26±6.04**	87.22±6.95	88.12±3.84	90.60±5.02
2	**89.64±3.91**	80.12±7.72	86.97±7.28	60.78±11.58	77.83±11.63	57.63±12.72	86.13±7.42	88.61±9.68	81.51±8.34	88.16±6.09
3	92.82±8.87	97.78±2.47	98.92±1.45	96.75±5.03	97.66±2.98	96.73±7.10	99.60±0.51	99.93±0.22	99.80±0.34	**99.94±0.12**
4	**94.64±2.37**	89.23±3.80	92.05±1.61	84.69±6.24	87.45±6.32	86.24±6.41	91.35±1.11	93.72±2.59	90.95±3.33	93.21±2.45
5	94.02±2.67	96.67±2.48	90.00±5.56	87.04±7.51	96.04±3.95	83.94±14.73	98.88±1.23	91.88±8.86	97.90±5.17	**99.74±0.75**
6	87.49±7.54	78.00±7.57	84.19±7.91	87.53±4.72	75.38±10.41	**91.22±5.75**	77.69±10.67	90.56±6.77	86.22±8.89	90.19±7.06
7	75.85±5.15	60.19±7.44	69.68±11.86	75.04±15.26	60.86±9.96	**82.18±10.41**	52.98±12.94	70.08±10.38	81.63±12.62	78.67±9.68
8	**54.85±13.38**	41.04±10.72	47.13±9.63	50.22±16.36	47.09±10.68	39.56±14.93	41.99±10.24	42.15±8.26	37.59±8.19	46.36±12.15
9	76.67±6.30	61.03±10.45	67.75±12.89	62.29±10.44	59.97±9.14	68.00±11.83	59.33±10.96	60.78±14.01	78.52±5.82	**80.23±5.46**
10	61.99±7.89	57.94±11.36	60.00±9.79	63.98±11.06	56.64±12.41	**68.74±14.49**	57.42±12.06	57.20±7.30	56.41±12.76	62.50±10.54
11	62.01±12.64	54.84±12.26	59.57±13.04	**81.68±12.15**	52.92±15.43	76.76±13.84	66.25±12.29	78.06±12.29	70.13±20.42	79.98±16.77
12	65.14±7.32	**79.94±7.44**	74.28±8.15	68.98±14.10	71.44±14.71	73.61±7.19	79.87±9.18	54.24±9.61	57.96±7.53	74.99±8.19
13	61.11±8.68	88.21±5.48	92.41±2.58	85.39±7.28	87.33±8.04	87.87±5.35	46.49±13.12	**94.22±2.14**	86.47±5.99	93.30±1.90
14	92.00±7.12	90.83±6.16	96.27±6.34	98.91±2.60	91.04±8.13	97.97±4.49	96.15±2.15	**100.00±0.00**	99.74±0.70	**100.00±0.00**
15	97.99±1.10	95.50±3.44	97.15±3.75	90.31±14.57	93.54±3.89	97.30±5.65	93.85±9.54	**100.00±0.00**	99.59±1.24	99.92±0.18
OA(%)	78.73±1.89	74.59±1.82	77.46±2.82	74.58±2.73	73.58±2.07	75.48±2.12	74.69±2.12	76.76±1.65	77.81±2.44	**82.56±2.25**
AA(%)	79.79±1.58	77.38±1.70	80.47±2.52	77.89±2.85	76.34±1.74	79.07±2.05	75.95±1.67	80.58±1.29	80.84±1.83	**85.19±1.75**
κ	0.77±0.02	0.73±0.02	0.76±0.03	0.72±0.03	0.72±0.02	0.73±0.02	0.73±0.02	0.75±0.02	0.76±0.03	**0.81±0.02**

misclassification stripes in the GF dataset. KSN outperforms most competitors by closely following natural land contours and avoiding the separation of land feature edges into a distinct category (observed in CMFSL and Gia-CFSL, especially in HU). In fine detail classification, KSN segments fine objects accurately in the HU dataset but faces challenges with blurring in the IP datasets due to increased patch size.

E. Impact of Training Sample Proportion

To assess robustness under varying training sample sizes, we evaluated the eight methods with additional few-shot settings (10, 15, and 20). For classes with limited data, we employed a bisected split (e.g., 22 samples split as {5,17} for five shots, {10,12} for ten shots, and {11,11} for both 15 and 20 shots).

Fig. 13 illustrates a clear upward trend in performance for each method as the number of training samples increases, evaluated using OA, AA, and κ. The proposed KSN consistently outperforms others on the IP and HU datasets in terms of OA and κ, while it leads in AA across all four datasets. For other methods, the cross-domain method by Lee et al. displays general middle performance across various datasets, with top-tier OAs in the GF dataset. Two FSL methods—S3Net and SPFormer—demonstrate performances similar to cross-domain methods. On the IP dataset, SPFormer and S3Net ranked second and third in OAs. However, their results on the GF dataset were less impressive. The remaining CDFSL methods, based on episodic training, demonstrated generally good performance, with HFSL and FSCF-SSL achieving top OAs on PU. However, their performance on the other three datasets was weaker.

Consequently, the results suggest that although KSN does not invariably achieve the highest accuracy, it consistently ranks as a leading contender.

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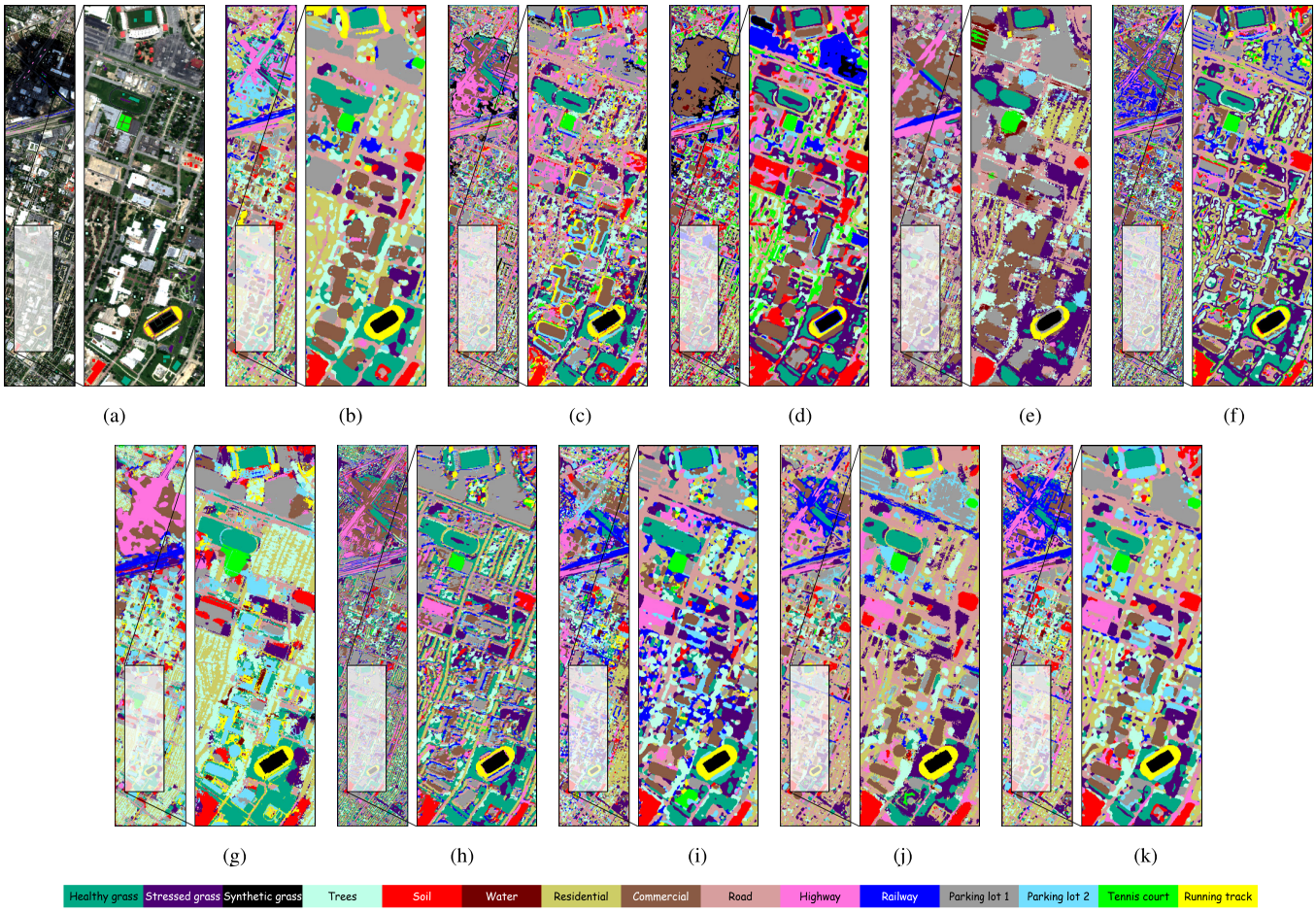


Fig. 11. Classification maps of HU with different methods: (a) ground truth, (b) Lee et al. (OA = 80.90%), (c) DCFSL (OA = 62.38%), (d) CMFSL (OA = 61.61%), (e) HFSL (OA = 69.81%), (f) Gia-CFSL (OA = 82.82%), (g) FSCF-SSL (OA = 74.80%), (h) ADAFSL (OA = 72.17%), (i) S3Net (OA = 80.72%), (j) SPFormer (OA = 80.05%), and (k) KSN (OA = 85.91%).

TABLE XI
CLASSIFICATION RESULTS ON THE GF DATASET, WITH FIVE SAMPLES PER CLASS

Class	Lee et al. [47]	DCFSL [38]	CMFSL [48]	HFSL [56]	Gia-CFSL [57]	FSCF-SSL [18]	ADAFSL [36]	S3Net [52]	SPFormer [58]	KSN (Ours)
1	**89.54±3.08**	80.90±6.81	81.13±5.02	81.91±7.74	79.95±6.80	84.02±6.41	81.96±7.12	76.57±3.93	80.75±5.12	83.63±6.25
2	97.58±2.06	97.04±4.66	98.18±3.52	99.32±0.92	98.52±3.27	**99.81±0.38**	99.22±1.07	60.03±14.70	92.49±12.96	97.93±3.98
3	85.00±4.93	86.48±11.89	85.62±15.19	86.19±12.03	89.72±10.10	77.90±20.72	88.29±12.91	84.38±13.60	82.86±18.24	**91.81±11.17**
4	90.64±3.25	**90.81±3.63**	85.68±6.63	80.11±12.05	87.67±5.41	76.00±15.14	88.38±2.73	79.51±10.82	79.08±7.70	88.54±4.12
5	88.09±9.04	84.87±10.79	93.72±8.74	93.08±4.06	79.87±8.87	91.54±2.51	79.49±13.47	89.23±9.78	91.41±7.45	**94.62±6.22**
6	99.29±1.30	98.13±2.65	95.28±4.79	83.19±12.70	94.40±5.74	81.98±14.29	97.80±1.90	98.57±2.36	97.47±4.51	**99.78±0.66**
7	93.53±6.10	93.11±5.28	96.78±4.78	87.33±8.14	87.00±6.99	87.11±10.27	91.56±6.92	92.78±4.69	**99.44±1.14**	96.22±5.52
8	99.56±0.56	100.00±0.00	98.79±1.43	94.17±6.15	99.61±0.52	94.90±5.03	99.56±0.67	91.80±8.93	95.53±6.10	**100.00±0.00**
9	98.01±2.01	96.31±2.59	**100.00±0.00**	97.85±2.77	97.69±2.78	98.51±2.45	97.80±2.28	**100.00±0.00**	**100.00±0.00**	99.59±0.82
10	89.58±4.88	82.10±13.68	79.73±17.40	**91.68±5.60**	79.94±14.79	87.85±6.04	85.65±8.92	70.32±12.89	62.60±14.11	86.47±8.75
11	98.72±1.28	97.94±2.86	**98.92±1.22**	93.87±6.04	96.24±3.38	96.41±3.57	96.95±6.23	96.50±5.31	93.38±6.79	97.34±7.69
12	99.10±1.41	**100.00±0.00**	99.98±0.07	97.41±4.37	99.94±0.19	98.99±2.09	99.85±0.45	99.63±1.10	99.85±0.39	**100.00±0.00**
13	98.05±1.87	**99.47±0.68**	99.33±1.01	97.41±4.80	97.78±2.85	96.97±4.01	96.18±2.84	98.57±1.61	97.84±3.09	99.02±1.67
14	**83.60±3.61**	76.04±5.25	76.85±7.93	55.79±17.97	77.36±6.50	56.60±12.86	78.08±6.56	54.72±13.93	68.38±12.14	76.13±7.64
15	99.39±0.43	99.54±0.52	99.69±0.54	98.32±2.18	99.78±0.41	**99.93±0.16**	98.52±1.38	93.20±4.77	95.19±3.29	99.42±0.98
16	97.53±1.49	**98.98±1.40**	98.58±1.70	90.73±7.69	97.97±3.32	93.47±7.10	97.19±3.82	94.52±3.72	93.12±4.57	96.17±5.89
17	**87.00±4.83**	82.42±9.96	85.08±11.13	76.56±12.55	84.30±10.04	75.31±8.78	85.00±10.95	83.83±11.28	79.14±11.84	84.37±11.22
18	77.18±11.10	69.11±17.20	73.04±17.62	64.17±24.37	68.52±16.53	73.04±18.04	71.86±14.14	69.30±17.55	72.61±18.53	**81.48±15.17**
19	74.80±4.65	90.64±8.74	90.79±8.06	86.51±11.21	92.06±8.81	86.98±13.35	89.68±7.79	92.70±4.98	**93.33±5.80**	92.86±6.74
20	84.08±8.13	86.27±10.44	87.46±8.78	**93.43±4.91**	84.18±14.16	92.54±4.95	87.46±7.76	87.76±10.98	92.54±4.72	91.64±5.00
21	94.21±6.85	95.88±5.76	99.41±1.18	99.41±1.18	94.12±5.26	**100.00±0.00**	97.35±3.82	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA(%)	**93.46±1.17**	91.31±2.22	91.56±2.25	89.96±1.74	90.65±1.90	90.51±1.44	91.87±1.66	82.48±2.14	86.27±4.17	93.12±1.05
AA(%)	91.64±1.14	90.76±1.40	91.62±1.50	88.02±2.42	89.84±1.52	88.09±2.03	90.85±1.35	86.38±2.02	88.91±2.38	**93.19±0.87**
κ	**0.93±0.01**	0.91±0.02	0.91±0.02	0.89±0.02	0.90±0.02	0.90±0.02	0.91±0.02	0.81±0.02	0.85±0.04	**0.93±0.01**

V. DISCUSSION

A. Do We Need More Than Spectral Shift Mitigation?

To assess the necessity of our work, we conducted seven experiments using combinations of SRC, original spectral data (RAW), and PCA-transformed features, as detailed in Table XII. Within these experiments, PCA is used to generate heterogeneous features, producing PCA_S and PCA_T for the source and TDs, respectively; SRC creates homogeneous features, maintaining spectral consistency across the

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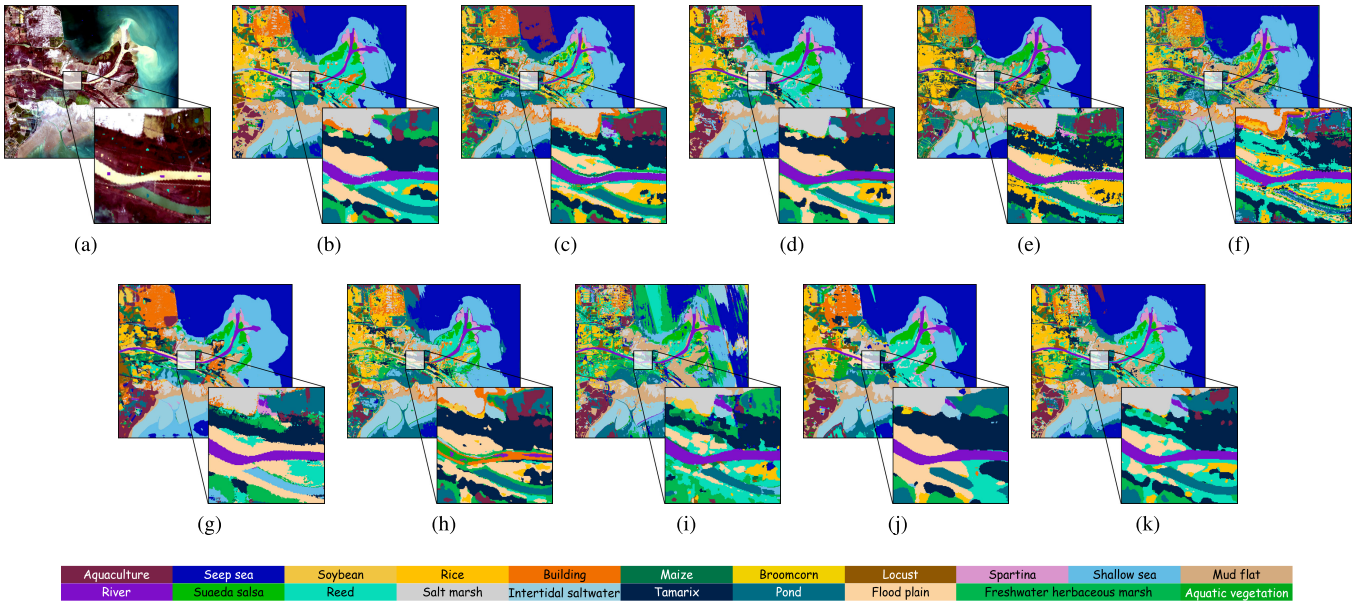


Fig. 12. Classification maps of GF with different methods: (a) ground truth, (b) Lee et al. (OA = 89.72%), (c) DCFSL (OA = 87.11%), (d) CMFSL (OA = 91.33%), (e) HFSL (OA = 90.45%), (f) Gia-CFSL (OA = 90.43%), (g) FSCF-SSL (OA = 90.79%), (h) ADAFSL (OA = 85.58%), (i) S3Net (OA = 80.42%), (j) SPFormer (OA = 86.46%), and (k) KSN (OA = 92.58%).

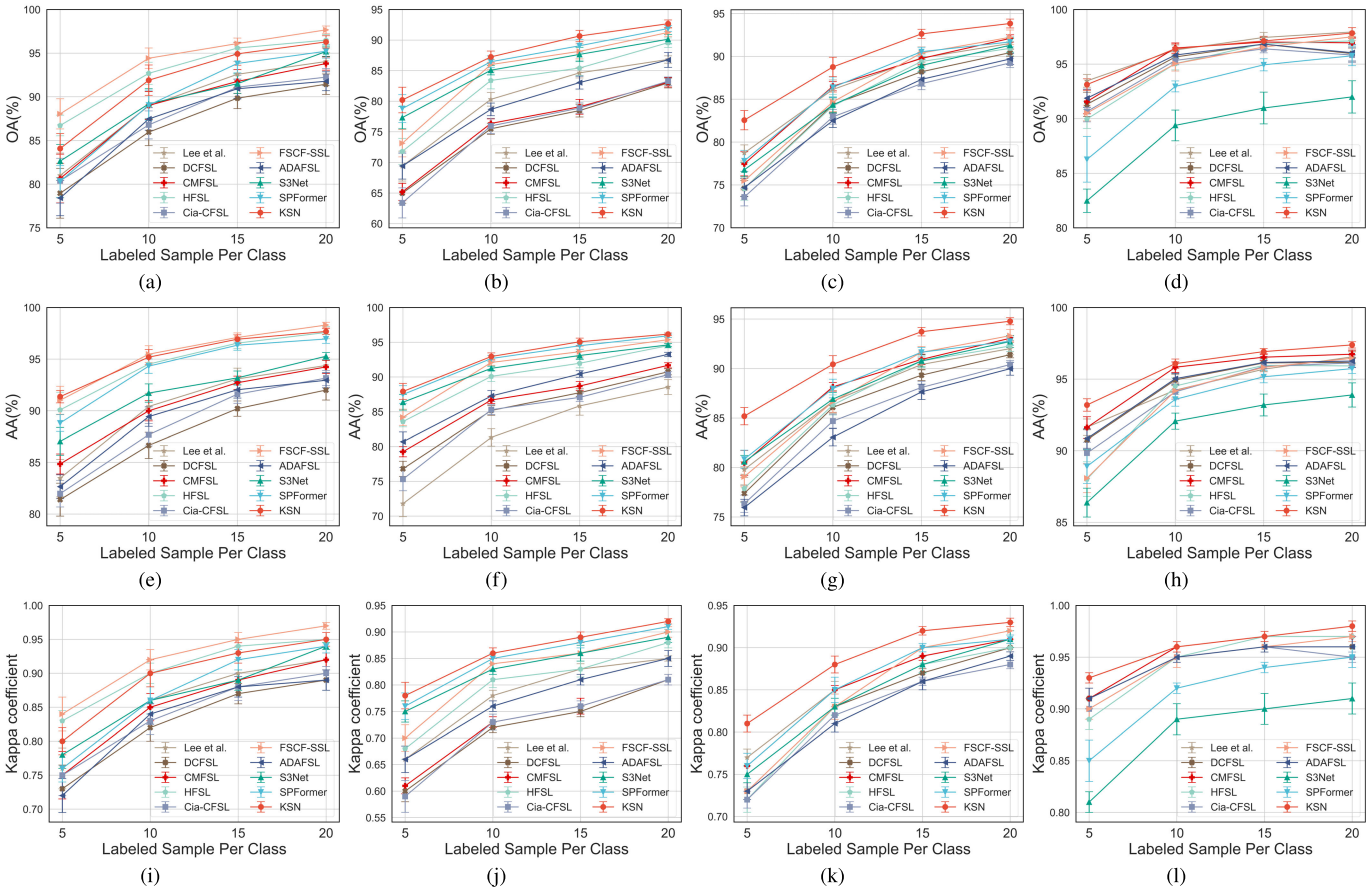


Fig. 13. Classification performance of different methods on four TD datasets with varying amounts of training samples. (a) OAs on PU dataset. (b) OAs on IP dataset. (c) OAs on HU dataset. (d) OAs on GF dataset. (e) AAs on PU dataset. (f) AAs on IP dataset. (g) AAs on HU dataset. (h) AAs on GF dataset. (i) κ on PU dataset. (j) κ on IP dataset. (k) κ on HU dataset. (l) κ on GF dataset.

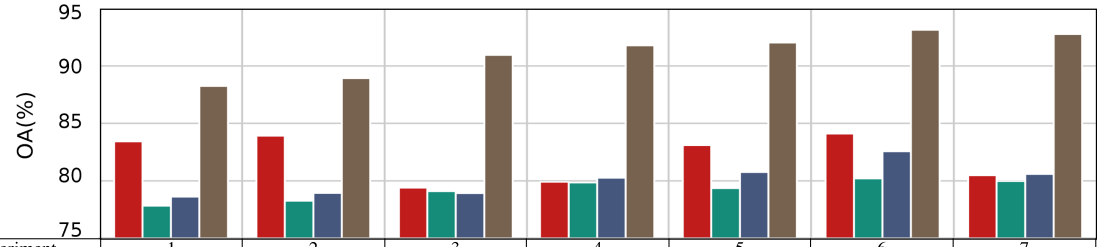
domains; RAW serves as another candidate for heterogeneous features.

Specifically, Experiment 2 represents the scenario where the spectral shift is neglected, Experiment 4 represents awareness of the specific shift but neglects TD-specific information, and Experiment 6 is the proposed KSN. Simultaneously, the versions excluding pretraining of these cases are presented in Experiments 1, 3, and 5, respectively.

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TABLE XII

CLASSIFICATION RESULTS OF KSN USING DIFFERENT FEATURE COMBINATION, WITH VALUES IN BRACKETS INDICATING THE DIFFERENCE FROM THE VERSION EXCLUDING PRETRAINING (W/O PRETRAIN)



Experiment		1	2	3	4	5	6	7
Feature	f_{θ_P}	w/o Pretrain	PCA_S	w/o Pretrain	SRC	w/o Pretrain	SRC	SRC
	f_{θ_T}	PCA_T	PCA_T	SRC	SRC	SRC	SRC	SRC
	f_{θ_S}	PCA_T	PCA_T	SRC	SRC	PCA_T	PCA_T	RAW
OA(%)	PU	83.42±4.16	83.91±3.46 (+0.49)	79.40±3.44	79.90±3.54 (+0.50)	83.09±3.08	84.09±3.38 (+1.00)	80.47±4.28
	IP	77.82±3.71	78.26±3.72 (+0.44)	79.09±4.28	79.84±4.12 (+0.75)	79.34±3.27	80.20±4.19 (+0.86)	79.96±3.58
	HU	78.60±1.64	78.93±1.42 (+0.33)	78.92±2.25	80.25 (+1.31)	80.75±2.30	82.56±2.25 (+1.81)	80.58±1.95
	GF	88.24±3.15	88.92±1.96 (+0.68)	90.94±1.70	91.77 (+0.83)	92.02±1.31	93.12±1.05 (+1.10)	92.75±0.98

Experiment 7 is an additional experiment for further exploration.

1) Effectiveness of Spectral Response Sampling (SRC): From comparisons of Experiments 1 versus 2 and 3 versus 4, an increase in enhancements from pretraining is observed. For the PU, IP, HU, and GF datasets, enhancements rose from 0.49% to 0.50%, 0.44% to 0.75%, 0.33% to 1.31%, and 0.68% to 0.83%, respectively. These results demonstrate that feature alignment via SRC effectively reduces spectral shifts across domains, thereby enhancing the benefits derived from pretrained weights. In addition, the comparison of Experiments 5 and 6 also highlights the benefits of homogeneous pretraining.

However, a significant performance decrease was observed in the PU dataset in Experiments 3 versus 4 compared to Experiments 1 versus 2, accompanied by a narrower gap in pretraining enhancement (from 0.49% to 0.50%). These reductions may be attributed to two reasons. 1) The loss of information caused by SRC, which significantly reduced the separability of PU features (as shown in Experiments 1 versus 3), and the loss could not be fully offset by transferring source knowledge alone (as seen in Experiment 4). 2) The high homogeneity between the source (PC) and target (PU) data (see Table III), which led domain-specific PCA to generate two highly informative yet relatively homogeneous feature spaces. Therefore, comparable enhancements can be observed in PU in Experiments 2 and 4.

2) Effectiveness of Introducing Heterogeneous Feature: The comparison between Experiments 4 and 6 shows that enhancements are more significant with the assistance of heterogeneous features. Furthermore, comparisons among Experiments 1, 3, and 5 suggest that, even without pretraining, KSN exceeds the performance of knowledge swap with solely heterogeneous or homogeneous features in most cases, thereby reinforcing the value of incorporating heterogeneous features.

Experiment 7 was conducted to further demonstrate that effectiveness does not depend on specific feature processing methods, such as PCA. Experiment 7, using original spectral data, achieved the best results among Experiments 2, 4, and 7, with the sole exception being PU in Experiment 2.

B. Ablation Analysis

To assess the individual and combined contributions of different components within the KSN, we conducted ablation studies. We systematically removed components from the S2RN backbone (SE-Res and RDD loss) and the knowledge swap strategy (Homo-KD and Hetero-ML). It is important to note that Hetero-ML, which leverages domain-specific feedback, depends on Homo-KD for initial spectral knowledge transfer. Consequently, we present ablation results for eight KSN variants in Table XIII. The specific analyses are as follows.

1) SE-Res: SE-Res enhances the spectral learning capability of the S2RN backbone by increasing model complexity and refining feature representation within the spectral module. To evaluate the impact of SE-Res, we compared variants V1 and V2. SE-Res consistently improved OA across all four datasets, with the most significant gain on the HU dataset (0.62%). Notably, combining SE-Res with RDD loss (V4) yielded the best performance overall. However, a slight decrease in OA (−0.05%) was observed on the IP dataset when comparing V3 and V4, suggesting that the benefits of SE-Res can be mitigated by the strong discriminative power provided by RDD loss in certain cases. Overall, SE-Res demonstrates a positive impact on model performance, although its effectiveness may vary depending on the dataset's inherent feature discriminability.

2) RDD Loss: RDD loss aims to improve feature discriminability, especially when dealing with limited samples, by leveraging intrabatch contrastive learning. Evaluating its impact by comparing V1 to V3 reveals significant OA improvements on the PU (2.15%) and IP (0.82%) datasets, along with a measurable gain on GF (0.29%). These observations hold true even when combined with SE-Res (V2 versus

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TABLE XIII
ABLATION COMPARISON OF EACH VARIANT OF KSN IN TERMS OF OA (%)

Variant	SE-Res	RDD loss	Homo-KD	Hetero-ML	PU	IP	HU	GF
V1	-	-	-	-	77.32±4.76	79.20±4.46	80.29±1.97	91.44±0.83
V2	✓	-	-	-	77.35±4.23	79.58±4.43	80.91±1.95	91.64±1.19
V3	-	✓	-	-	79.47±4.15	80.02±4.08	80.31±2.43	91.73±1.33
V4	✓	✓	-	-	79.60±4.16	79.97±4.12	80.92±2.22	91.85±1.43
V5	-	-	✓	-	82.43±4.21	79.27±3.83	82.38±2.38	92.95±1.40
V6	-	-	✓	✓	82.30±4.17	79.49±3.86	82.20±2.09	92.85±1.24
V7	✓	✓	✓	-	83.87±3.61	80.05±4.19	82.25±2.33	93.02±1.21
KSN	✓	✓	✓	✓	84.09±3.38	80.20±4.19	82.56±2.25	93.12±1.05

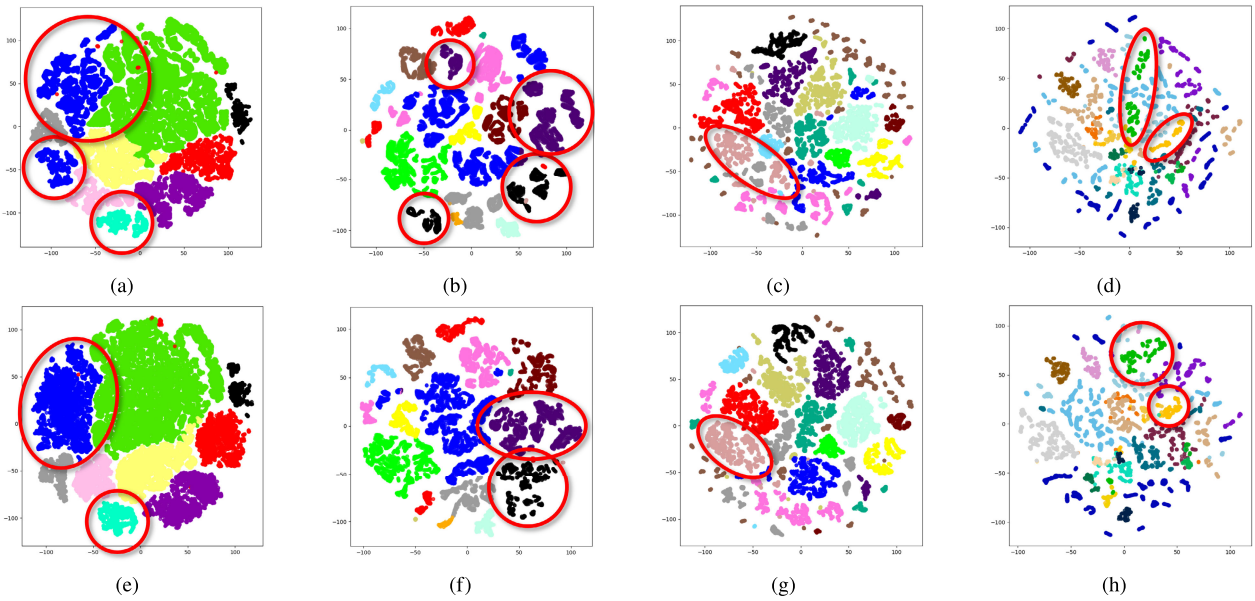


Fig. 14. t-SNE visualization (perplexity = 15) of embedding features extracted by KSN for four hyperspectral datasets (PU, IP, HU, and GF). The top row shows features without RDD loss, while the bottom row includes RDD loss. Red circles highlight regions where RDD loss improves class separability by reducing the occurrence of single classes forming multiple clusters in the t-SNE space. (a) PU w/o RDD loss. (b) IP w/o RDD loss. (c) HU w/o RDD loss. (d) GF w/o RDD loss. (e) PU w/RDD loss. (f) IP w/RDD loss. (g) HU w/RDD loss. (h) GF w/RDD loss.

V4). In line with these findings, the t-SNE visualization in Fig. 14 demonstrates the effectiveness of RDD loss in enhancing feature discriminability. Samples within classes exhibit improved concentration, as highlighted by red circles. The lack of significant impact on the HU dataset can be attributed to its inherent high spectral resolution and diverse land cover. These characteristics contribute to better class separation [see Fig. 14(c)], diminishing the need for RDD loss's additional discriminative power.

3) Homo-KD: Homo-KD, a critical step in the knowledge swap strategy, facilitates the transfer of homogeneous model-inherited SD knowledge and the learned TD knowledge to heterogeneous models. The ablation results comparing V1–V5 and V4–V7 demonstrate the significance of Homo-KD. It led to substantial accuracy improvements on the PU dataset (4%–5%) and notable gains on the HU (1.6%) and GF (1.4%) datasets. These results underscore Homo-KD's crucial role in KSN. However, the IP dataset exhibited minimal improvement (less than 0.1%), which is likely due to the choice of the SA dataset as the SD. Although SA shares similar sensor attributes, its distinct spatial resolution and land cover create a nonspectral domain shift, hindering the effectiveness of Homo-KD. The superior performance of V7 compared to V5 and V4 reinforces the synergistic relationship between SE-Res, RDD loss, and Homo-KD within the KSN framework.

4) Hetero-ML: Hetero-ML refines the homogeneous model by feeding back the performance of the heterogeneous model on a small validation set, influencing the homogeneous model's parameter updates to better capture TD-specific features. Comparing V7 and V8, we observed consistent accuracy improvements across all datasets when combining Hetero-ML with SE-Res, RDD loss, and Homo-KD (0.22% on PU, 0.15% on IP, 0.31% on GF, and 0.10% on HU). However, when used with only Homo-KD (V5 versus V6), improvements were limited to the IP dataset. The outcome can be attributed to the reliance of the heterogeneous model on pseudo-labels generated by the homogeneous model. Consequently, enhancing the homogeneous model's performance is essential for the heterogeneous model to provide accurate feedback, leading to overall improvements. This observation underscores the synergistic relationship between the four components of KSN.

C. Complexity Analysis

As shown in Table XIV, we present a comparative analysis of parameter size (Para.), multiply-accumulate operations (MACs), and training/testing durations for different methods. Due to the reusability of pretrained models, pretraining times for KSN, Lee et al., HFSL, FSCF-SSL, and SPFormer are omitted. For reference, KSN pretraining on the PC dataset took an average of 2.34 s/epoch.

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TABLE XIV
PARAMETER SIZE, MULTIPLY-ACCUMULATE OPERATIONS, TRAIN, AND TEST. DURATIONS OF DIFFERENT METHODS

Dataset	Metric	Lee et al. [47]	DCFSL [38]	CMFSL [48]	HFSL [56]	Gia-CFSL [57]	FSCF-SSL [18]	ADAFSL [36]	S3Net [52]	SPFormer [58]	KSN (Ours)
PU	Para.	301.71 k	59.87 k	614.01 k	3.16 M	208.69 k	3.16 M	49.24 k	91.52 k	41.9 k	330.08 k
	MACs	16.27 M	42.49 M	27.15 M	201.13 M	42.64 M	200.26 M	29.99 M	63.34 M	5.44 M	198.96 M
	Train.	10.10 s	721.88 s	460.62 s	107.62 s	1672.68 s	391.24 s	1367.13 s	5.94 s	4.30 s	29.50 s
	Test.	5.41 s	3.63 s	1.85 s	1.41 s	3.50 s	11.48 s	4.98 s	9.56 s	7.16 s	15.09 s
IP	Para.	386.7 k	70.70 k	614.02 k	3.30 M	208.70 k	3.3 M	60.06 k	155.01 k	78.26 k	579.59 k
	MACs	17.35 M	42.49 M	27.16 M	200.26 M	42.65 M	201.13 M	30.78 M	120.1 M	14.47 M	273.99 M
	Train.	14.30 s	1212.02 s	783.32 s	2850.41 s	2862.67 s	570.83	2255.81	12.34 s	6.77 s	66.46 s
	Test.	1.30 s	1.18 s	0.50 s	3.94 s	0.97 s	4.23 s	1.35 s	2.87 s	2.11 s	4.05 s
HU	Para.	369.17 k	64.94 k	614.03 k	3.22 M	208.71 k	3.22 M	54.3 k	93.28 k	42.09 k	430.98 k
	MACs	16.75 M	42.49 M	27.17 M	200.62 M	42.66 M	200.62 M	30.33 M	30.65 M	5.6 M	98.21 M
	Train.	12.56 s	1043.53 s	412.81 s	162.11 s	2589.80 s	480.55 s	2040.48 s	8.85 s	9.03 s	26.97 s
	Test.	1.61 s	1.78 s	0.75 s	4.52 s	1.35 s	4.85 s	2.66 s	3.10 s	3.22 s	3.63 s
GF	Para.	449.43 k	79.52 k	614.04 k	3.44 M	208.72 k	3.44 M	68.89 k	156.47 k	78.46 k	536.23 k
	MACs	18.28 M	42.49 M	27.18 M	201.91 M	42.67 M	201.91 M	31.47 M	34.97 M	10.77 M	142.54 M
	Train.	22.82 s	1445.60 s	454.51 s	323.09 s	3692.04 s	872.15 s	2550.17 s	17.91 s	6.97 s	41.97 s
	Test.	1.04 s	1.08 s	0.28 s	4.46 s	0.67 s	3.97 s	1.29 s	2.10 s	1.25 s	1.70 s

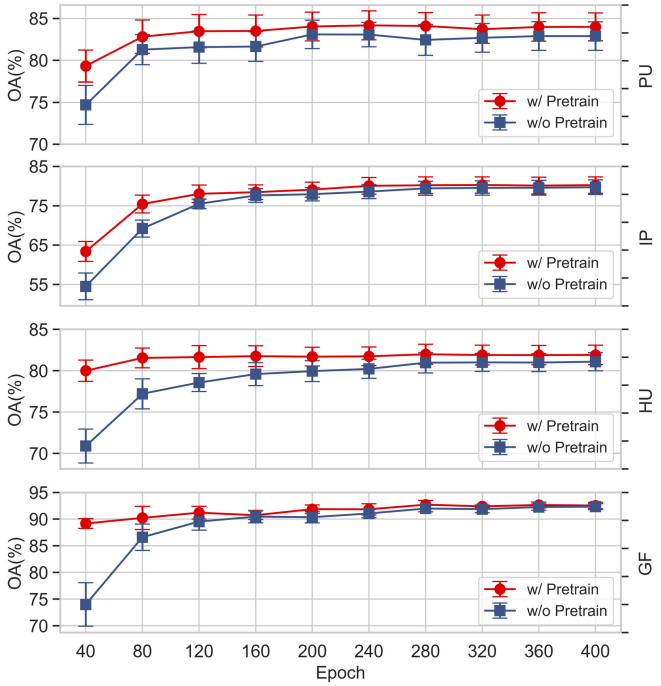


Fig. 15. Comparison of convergence speed of KSN with and without pretraining.

KSN exhibits a larger parameter count and higher MACs compared to other methods primarily due to the introduction of the heterogeneous model. However, this increased complexity is offset by faster convergence, enabling KSN to achieve the desired accuracy within 80–120 epochs compared to nearly 280 epochs for models initialized randomly (see Fig. 15). Consequently, the overall training time of KSN is comparable to or shorter than other methods.

Regarding testing time, the patch cropping strategy plays a crucial role. Methods like DCFSL using predefined patches are generally faster, while real-time cropping methods like ours incur additional computational overhead. Moreover, KSN's dual-model inference process contributes to slightly longer testing times.

Overall, the increased parameters and computational load of KSN have a minimal impact on computational efficiency.

VI. CONCLUSION

In this work, we propose the KSN as a novel approach for cross-domain few-shot HSI classification. KSN effectively addresses the challenges posed by spectral shifts by establishing a homogeneous feature space and employing a knowledge swap strategy that combines Homo-KD and Hetero-ML. Extensive experiments demonstrate the superior performance of KSN in terms of accuracy and robustness compared to state-of-the-art methods.

While KSN demonstrates promising results, its performance is influenced by the quality and representativeness of the source data. Limited and biased source data can lead to suboptimal performance, particularly when transferring to TDs with significantly different spectral characteristics. Future research will focus on enhancing the pretraining phase and exploring techniques to improve the method's adaptability to diverse TDs. In addition, investigating the potential of multiple SD pretraining or meta-learning for KSN could be a valuable direction for future work.

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